A Mini Project with Seminar

ON

DIABATES SEVERITY PREDICTION USING RETINAL FUNDUS IMAGES

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Computer Science and Engineering (Data science)

by

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DECLARATION

We hereby declare that the major project titled **"Diabetes Severity Prediction using Retinal Fundus Images"** is the work done during the period from **16th January 2026 to 23th April 2026** and is submitted in the partial fulfillment of the requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering (Data science) from Gokaraju Rangaraju Institute of Engineering and Technology (Autonomous under Jawaharlal Nehru Technology University, Hyderabad). The results embodied in this project have not been submitted to any other University or Institution for the award of any degree or diploma.

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ABSTRACT

Diabetes is a long-term metabolic condition that is typified by high levels of glucose in the blood, which when not managed results in serious complications such as diabetic retinopathy, kidney failure and heart failure. A microvascular complication of diabetes, diabetic retinopathy (DR) is an irreversible condition that involves the micro vessels of the retina and is directly proportional to long-term glycemic control, rendering the retina a particularly attractive non-invasive measure of systemic diabetic severity. Traditional diabetes diagnosis is based on invasive blood tests and lab facilities, which are not always accessible in resource-restricted environments, and there is a strong necessity of non-invasive, imaging-based diagnostic options. This project suggests a five-stage end-to-end pipeline to predict the severity of diabetes by relying on retinal fundus images and non-invasively determining the level of fasting blood glucose in the body based on a single fundus image. The first stage is preprocessing retinal images with Ben Graham illumination correction and CLAHE, followed by segmentation of pixel lesions with a U-Net encoder based on EfficientNet-B4 trained on the FGADR dataset, with combined Dice and Focal loss, with a best mIoU of 0.1116. A ResNet-50 CNN is then used to classify the severity of DR into five levels based on a 4-channel input with three RGB channels and the U-Net lesion mask generated, with a validation accuracy of 88.09%. The DR grade is predicted and mapped to an HbA1c range through UKPDS clinical lookup tables, the mean HbA1c is extracted and used in combination with DR grade to create a 7-feature vector. The four regression models of ADA Formula, Lasso Regression, CART and random forest are trained and compared on NHANES paired HbA1c-glucose data, where Lasso Regression performs best with an MAE of 10.48 mg/dL and R 2 of 0.6283 and is chosen as the final glucose prediction model. The suggested system illustrates the practicability of non-invasive estimations of blood glucose to retinal images and has a high potential of performing extensive screening of diabetic patients in healthcare settings that are limited in terms of resources.

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LIST OF ACRONYMS

Acronym	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
U-Net	U-Shaped convolutional network
ResNet	Residual Neural Network
DR	Diabetic Retinopathy
IRMA	Intraretinal Microvascular Abnormalities
Hb1AC	Hemoglobin A1c
NHANES	National Health and Nutrition Examination Survey
CLAHE	Contrast Limited Adaptive Histogram Equalization
RGB	Red, Green, Blue
CART	Classification and Regression Tree
GPU	Graphics Processing Unit
CPU	Central Processing Unit
MSE	Mean Squared Error
MAE	Mean Absolute Error
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
DSC	Dice Similarity Coefficient
IoU	Intersection over Union
ReLU	Rectified Linear Unit
API	Application Programming Interface
GUI	Graphical User Interface
IDE	Integrated Development Environment
FP	False Positive
FN	False Negativity
TP	True Positive
TN	True Negative

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CHAPTER 1
INTRODUCTION

1.1 Background and Context

One of the most common long-term illnesses that afflict millions of people worldwide is diabetes mellitus. It is a metabolic disorder that occurs when the body is unable to produce sufficient insulin or cannot effectively utilize the insulin produced. Consequently, this causes the level of blood glucose to be high, which causes chronic health problems. These complications involve cardiovascular diseases, kidney damage, nerve disorders and impairment of vision. The rising incidence of diabetes has emerged as a significant issue to health care systems in all parts of the world. The disease can be controlled with early diagnosis and constant monitoring. The patients are vulnerable to severe complications without the necessary monitoring. As such, effective and consistent diagnostic methods are required to enhance patient outcomes and quality of life.

Traditional tools of measuring the level of glucose in the blood are mostly invasive. Finger-prick tests and blood analysis in the lab are some of the techniques that need patients to be drawn to provide a sample of the blood. Although these approaches are regarded as accurate and reliable, they might be uncomfortable and inconvenient to use on a regular basis. Repeated testing can be painful, uncomfortable and anxious particularly in older patients and children. This tends to decrease the compliance of the patients, which influences the management of the disease. Also, the laboratory tests can be time-consuming, expensive, and require medical facilities. These drawbacks underscore the necessity of other methods which are not as invasive and user friendly. Therefore, non-invasive methods that will allow monitoring glucose levels continuously and painlessly are in high demand.

As the field of artificial intelligence and medical imaging rapidly develops, it has become possible to find new opportunities in non-invasive disease diagnosis. Retinal fundus imaging especially has acquired a lot of importance as it has been able to provide a detailed image of blood vessels. The retina is a window through which to view vascular alterations with systemic illness, such as diabetes. A frequent complication of diabetes is Diabetic Retinopathy, which results in structural alterations in the retina including microaneurysms, hemorrhages and exudates. Such alterations are identifiable using retinal pictures and can be assessed in regards to the

severity of the disease. With

the help of these features, one can learn about the development of diabetes. Retinal imaging would, therefore, be a good source of information in non-invasive diagnosis and monitoring.

Convolutional Neural Network and other deep learning methods have proven to be exceptional in the analysis of medical images. U-Net models have become popular in the segmentation of significant regions in medical images with high precision. In a similar way, other architectures such as ResNet are quite useful in classification, allowing the correct determination of the levels of disease severity. These models are able to learn complex patterns and features with large datasets automatically without human intervention. Integrating both segmentation and classification techniques, one is able to derive valuable information out of retinal images. Also, it is possible to use regression models in order to predict continuous variables like blood glucose levels. This combination of several methods is a robust system of smart healthcare solutions. In that regard, the proposed system will create a non-invasive approach to estimating the level of blood glucose with the help of retinal fundus images. The system uses the concept of preprocessing to improve the quality of the image and the visibility of the features. It subsequently applies deep learning models to detect lesions and classify diseases. According to the classification results, additional analysis is conducted to give glucose levels based on regression methods. The technique minimizes reliance on the conventional invasive techniques and improves the comfort of the patients. Also, it offers efficient and scalable early diagnosis and monitoring solutions. The system helps to develop AI-based healthcare technologies. In general, it will make the process of diabetes management more accessible, accurate, and user-friendly.

1.2 Problem Statement

Regular blood glucose level monitoring is a major aspect of managing diabetes that can be used to avoid serious complications. Nonetheless, the majority of the currently available methods of monitoring are invasive and demand blood samples. Some testing methods like finger-prick testing can be painful and inconvenient when done regularly, even though they are accurate. This discomfort can result in low patient compliance particularly in the elderly and children. Moreover, the repeated tests may lead to stress and anxiety, which can create problems with regular monitoring practices among the patients. Consequently, most patients do not monitor their glucose levels. This heightens the chances of uncontrolled diabetes and other health complications. Thus,

alternative approaches that are less invasive and more comfortable to the patients are urgently required.

The other significant drawback of the existing diagnostic methods is that they require the presence of clinical settings and special equipment. The tests done in a laboratory must be carried out by trained staff, the infrastructure is necessary and time is needed to analyze them which is not always available to everyone. In rural or low-resource settings, the facilities may not be easily accessed and regular monitoring may not be feasible. Moreover, currently available solutions lack real-time and constant monitoring of the condition of a patient. Such a lag in the diagnosis and monitoring may result in late-complications of the disease including diabetic retinopathy. Therefore, an accurate, accessible, cost-effective, and timely-result-delivering solution is required.

Despite positive outcomes of medical imaging and artificial intelligence, the gap between the development of a fully integrated non-invasive system to estimate glucose still exists. The bulk of the existing literature is either the detection of diabetic retinopathy or the classification of the severity of the disease but not regarding the prediction of real blood glucose levels. Also, not all systems package segmentation, classification, and regression into a pipeline. This disintegration has restricted the overall performance of the existing solutions. Therefore, it is important to design a system that leverages retinal imaging along with advanced deep learning techniques. This system must have the ability to scan the retinal characteristics and match them to useful glucose level forecasts.

As a solution to these difficulties, the proposed work is aimed at creating a non-invasive, AI-based system to estimate the level of blood glucose by analyzing retinal fundus photos. The identified problem can be summarized as the necessity to substitute or decrease the reliance on intrusive glucose monitoring techniques with a smart and user-friendly one. The system is supposed to identify retinal abnormalities correctly, categorize the level of disease and predict glucose levels based on sophisticated models. It should also be reliable, efficient and easy to use in practical use. The proposed system will help in solving this problem to improve the management of diabetes and improve the quality of life of the patients. Finally, it leads to the creation of convenient and innovative healthcare systems.

1.3 Objectives of the Proposed Work

The main aim of the suggested work is to create a fully automated, non-invasive

computational pipeline which can accurately estimate the level of fasting blood glucose and predict the severity of diabetic retinopathy directly based on the retinal fundus images, without involving any type of blood sampling, lab testing or professional clinical intervention. This purpose is motivated by the increasing global disease burden of diabetes mellitus where millions of patients in the low- and middle-income countries cannot access routine glycemic testing because of the expense, infrastructure support, and invasive nature of the traditional blood glucose monitoring approaches. The proposed system will fill this gap by taking the clinical association between retinal microvascular findings and systemic glycemic status, and utilizing it to provide practical blood glucose information that can be actively used to track the severity of diabetes by using a simple, inexpensive, and universally applicable imaging-based method. The general intent is therefore not only academic but geared towards a practically implementable diagnostic tool to supplement primary care and diabetic screening interventions in resource strained healthcare settings.

Objectives:

- To preprocess retinal fundus images with black border removal, Ben Graham illumination correction, and CLAHE to improve image quality to feed subsequent deep learning models.
- On the FGADR dataset, the training of a U-Net with EfficientNet-B4 encoder with combined Dice and Focal loss to segment pixels at the pixel level into six retinal lesion categories viz. Microaneurysms, Hemorrhages, Hard Exudates, Soft Exudates, Neovascularization, and IRMA was conducted.
- To create a ResNet-50 CNN that accepts a 4-channel input of three RGB channels and the U-Net lesion mask to detect diabetic retinopathy severity with five grades ranging between Grade 0 and Grade 4.
- To implement a confidence thresholding mechanism on the CNN output with a threshold of 0.45 to identify cases where the DR grade prediction is uncertain and avoid unreliable grades being passed to subsequent steps.
- To interpolate the predicted DR grade to a clinically validated range of HbA1c to use in glucose feature engineering using UKPDS lookup tables and also retrieve the midpoint value of HbA1c.
- To create a 7-feature vector based on the extracted HbA1c midpoint and predicted DR grade based on the NHANES HbA1c-glucose dataset and feature engineer both linear and non-linear relationships.

- The training and comparison of four regression models namely ADA Formula, Lasso Regression, CART Decision Tree and Random Forest on the engineered feature table and the best performing model in terms of lowest MAE to be selected to use in final glucose prediction.
- To predict non-invasively the fasting level of blood glucose in mg/dL as the ultimate result of the proposed pipeline using a single retinal fundus image.

1.4 Scope of the Proposed Work

The proposed system of diabetes severity prediction based on retinal fundus images is scoped to cover the end-to-end computational pipeline where a raw retinal fundus image is the input and a predicted level of fasting blood glucose in mg/dL is the output. The system covers the entire process of preprocessing images, lesion classification, DR grade, HbA1c estimation, as well as glucose prediction, within its frame of reference. The system is to work with standard retinal fundus images and is trained and tested on the FGADR dataset only to the deep learning parts and the NHANES dataset only to the glucose prediction part.

The preprocessing pipeline has been designed to deal with typical retinal image quality problems like illumination or contrast disparities by using Ben Graham correction and CLAHE respectively. It does not involve processing of pathology image artifacts like extreme motion blur, camera lens flare or extreme underexposure problems that would necessitate the use of special image restoration algorithms that are outside the scope of this project. The system works on the images, which are downscaled to a common resolution of 512*512 pixels, and any image that cannot be meaningfully preprocessed along this pipeline is deemed to be out of scope of the proposed system.

The lesion segmentation part is bounded to identify and categorize six distinct lesion types that the FGADR dataset stipulates, such as Microaneurysms, Hemorrhages, Hard Exudates, Soft Exudates, Neovascularization and IRMA. It goes no further to detect other retinal diseases like diabetic macular edema, optic disc swelling, or retinal vascular occlusions that are clinically important but not included in the annotation of the FGADR dataset. The segmentation model generates a 7-class output mask with the background class and the localization of lesions is restricted to pixels only without any additional quantitative measurements of lesion size, shape and spatial distribution.

The DR grading component is scoped to classify retinal images into five severity

grades from Grade 0 to Grade 4 as per the International Clinical Diabetic Retinopathy severity

scale. The system has a built-in confidence thresholding mechanism of 0.45 that is within scope to trigger a low-confidence prediction, although there is no mechanism provided to request the clinical to input more information in a flagged case. The range of the DR grading phase is also narrowed down to single-eye analysis, i.e. the system works with one fundus image at a time and does not make bilateral comparisons between left and right eye images.

The biochemical estimation part is limited as it relies on the UKPDS clinical lookups to map DR grade to HbA1c ranges only, and does not consider other clinical biomarkers like fasting insulin levels, body mass index, or disease duration, which have been shown to affect blood glucose levels but cannot be obtained based on retinal images alone. The glucose prediction phase is scoped to a comparison of four particular models i.e. the ADA Formula baseline, Lasso Regression, CART Decision Tree and Random Forest Regressor, trained on a seven feature engineered table based on the NHANES data, where Lasso Regression is the best performing model based on the least MAE to make final prediction.

The suggested system is not associated with it under the scope of longitudinal tracking of patients, treatment advice, electronic health record, or real-time monitoring capability. It is confined to a single-visit, single-image based assessment of diabetes severity and blood glucose estimation which is a proof-of-concept non-invasive diagnostic aid that is expected to complement not to substitute clinical decision making by competent medical practitioners.

1.5 Methodology and Approach

The proposed methodology is a pipeline that is structured to process retinal fundus images to estimate the blood glucose level non-invasively. The system adheres to the sequential process which includes preprocessing, segmentation, classification and regression, as mentioned in Figure1. The stages have a particular role of converting raw input data to meaningful outputs. The methodology makes sure that relevant retinal features are successfully extracted and analyzed. It is structured in such a way that it is accurate and consistent at each stage. The general strategy is a combination of various deep learning methods to attain confident predictions. The workflow is arranged in a modular way to understand and apply it better. This hierarchical design enhances the performance and scalability of the system.

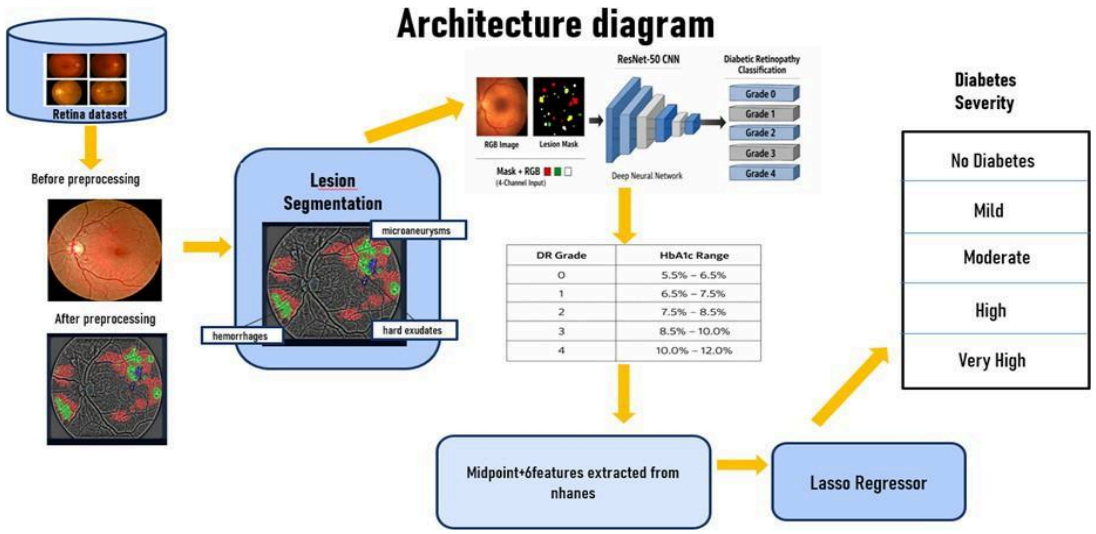


Figure 1 Architecture Diagram

Image preprocessing is the initial step in the methodology as it is aimed at improving the quality of retinal images. Methods like Ben Graham illumination correction are used to equalize the light conditions in pictures. Moreover, the Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to enhance the contrast and accentuate subtle features. The preprocessing will be used to minimize the noise and enhance the lesions visibility. The input images have to be of high quality to ensure that the model performs better. The step assures that the data presented to the following models is well-processed. It is also useful in managing the variations in the quality and conditions of image acquisition. Preprocessing, therefore, constitutes a very important basis of the whole pipeline.

The second step is the lesion segmentation with a U-Net architecture and an EfficientNet-B4 encoder. In this model, important retinal abnormalities (microaneurysms, hemorrhages, exudates) are detected at the level of pixels. The segmentation output indicates areas of interest to the retinal image. These divided features are important to comprehend disease patterns. The system enhances the quality of features in subsequent stages by the accurate detection of the lesions. U-Net is used to provide the accurate localization of abnormalities. EfficientNet increases the ability to extract features. This set of features is a robust segmentation model of medical images.

After the segmentation, the system classifies the severity of diabetic retinopathy with a ResNet-50 model. The segmented output is added to the original picture to give enhanced input to the classifier. The model groups images into various levels of severity of disease starting with no disease and up to advanced stages. This type of

classification

will assist in comprehending the evolution of diabetes in the patient. Proper classification is critical to make reliable glucose estimates. The ResNet-50 is selected due to the capacity to learn deep features and deal with intricate image patterns. The results of the classification are a valuable input of the subsequent step. The step increases the diagnostic strength of the system.

In the last step, the system approximates the level of blood glucose using regression methods. The levels of the severity that are classified are converted to the HbA1c values due to the established clinical correlations like the UKPDS model. The Decision Tree Regressor is then used to forecast the blood glucose levels in mg/dL. The step transforms qualitative outcomes of classification into quantitative forecasts. The regression model is established to give interpretable and consistent results. Clinical mapping enhances the predictability of prediction. This step accomplishes the conversion of image analysis to glucose estimation. In general, the methodology is a thorough and efficient method of non-invasive monitoring of diabetes.

1.6 Organization of the Report

The project report is presented in several chapters, devoted to a particular aspect of the proposed system.

Chapter 1: Gives an introduction to the problem, its background, problem statement, objectives, scope, and methodology of the work. It provides a clear picture on the motivation of the project and the method which is used to address the problem. This chapter is the basis of the whole report as it explains the significance of non-invasive glucose monitoring. It also lays emphasis on the important elements in the system. The introduction makes sure that the readers can easily understand the following chapters. Overall, it establishes the context and direction of the project.

Chapter 2: Depicts the literature review on current methods associated with the detection of diabetes and retinal image analysis. It is backed by research studies on different research papers, methodology, and techniques applied in the same fields. The chapter contrasts various methods in terms of their merits, demerits, datasets, and findings. To understand and compare, a summary table is also provided. Also, shortcomings of the available approaches are addressed in a clear way. This can be used to determine gaps in the research and to establish why the proposed system is necessary. The literature review is a key to the knowledge of the existing tendencies. It also aids in the formulation of a better solution.

Chapter 3: Explains the proposed methodology, and all the modules that will be part of the system. It describes how the preprocessing methods, segmentation models, classification methods and regression works. UML diagrams (use case, class, sequence and activity diagrams) are also provided in the chapter. There is also discussion of requirements engineering which comprises functional and non-functional requirements. Hards and soft requirements needed to be implemented are well defined. This chapter gives a full technical insight into the system design. It makes sure that the implementation process is clear. On the whole, it is the center of the project.

Chapter 4: Focuses on the results and discussions of the proposed system. It gives extensive details regarding the dataset utilized, its nature and structure. Results of the experiment are provided in the form of graphs, tables and sample outputs. The system performance is examined with the help of pertinent evaluation indicators. The benefits of the suggested solution are outlined. The limitations and challenges faced during the implementation are also discussed. This chapter assists in the comprehension of the effectiveness of the system. It gives a rationale to the methodology adopted. On the whole, it proves the practical results of the project.

Chapter 5: Will be the last part of the project where summarizing the main findings and contributions of the proposed work will be presented. It emphasises the significance of the system and its possible effect on healthcare. The conclusion is a reflection of the objectives and their achievement or not. Furthermore, it is proposed to make some improvements to the system in the future. It can involve real-time application, bigger datasets and sophisticated models. The chapter provides directions for future research and development. It makes sure that the scope of a project can be expanded. In general, it gives a sense of closure to the report.

Chapter 6: Contains appendices and references utilized in the project. Sample code snippets and other supporting material are given in the appendix section. All of the research papers, articles, and sources mentioned in the report are listed in the references section. Proper citation format is followed as per guidelines. This ensures academic integrity and avoids plagiarism. The chapter is a supplementing part of the whole document. It gives a plausibility to the presented work. In general, it finishes the paper work in a well-organized and professional way.

CHAPTER 2

LITERATURE SURVEY

2.1 Summary of Existing Approaches

Recent research has demonstrated great improvements in artificial intelligence application to detect and analyze diabetic retinopathy. Initial efforts of Saxena et al. [1] used the deep learning methods on retinal datasets of large scale including EyePACS and Messidor with high accuracy and sensitivity in the detection of retinal abnormalities. Abhishek et al. [2] introduced the lightweight CNN model that is built upon MobileNet that minimizes the computational needs and provides the possibility of performing the diagnosis in real-time, but with a slightly reduced accuracy. In a different research, deep learning features were combined with the conventional color and texture features to enhance classification among the various levels of severity of DR, in which focal loss was utilized to address the issue of class imbalance [3].Subsequently, deep learning models like ResNet and VGG were applied to automatically classify diabetic retinopathy with high accuracy and minimum human intervention, but large datasets were essential to train these models [4]. Other studies further utilized retinal imaging to forecast systemic health events such as cardiovascular diseases and hypertension with transfer learning, but still, interpretability of the model was an issue [5]. Segmentation and classification with CNNs yielded better results in lesion detection, revealing the presence of microaneurysms, hemorrhages and exudates, but demanded more calculations [6].U-Net-based segmentation methods and U-Net-based classification models, like ResNet, allowed detecting early signs, such as microaneurysms, with high accuracy, yet they are susceptible to noise and changes in light [7]. Combination of multiple models to form ensemble learning methods have also been suggested to enhance prediction accuracy but they increase the complexity of the system and training time [8]. Recent progress has involved using CNN with the Vision Mamba to extract local and global image features in an effective manner to enhance overall performance [9].

Model systems that combine models such as ResNet and DenseNet have demonstrated better stability and lower prediction error in various data sets, but are computationally costly to train [10]. Other studies were concerned with the prediction of diabetic retinopathy progression based on deep learning and time-series analysis, which offers insights into early intervention [11]. Other studies showed that there are

methods to

detect diabetes even before some retinal damage is noticeable by observing some latent patterns in retinal images, though generalization across datasets has been a problem [12]. Large-scale screening of diabetes has also been considered using retinal imaging as it is non-invasive, but with a slightly smaller accuracy than sophisticated techniques [13]. Moreover, the integration of conventional methods of image processing with deep learning has enhanced feature extraction by first improving the quality of the image prior to analysis but at a cost of complexity on the system [14]. The traditional methods, which rely on the digital image processing methods including segmentation and thresholding have also been investigated in detecting diabetic retinopathy [15]. These techniques are easy, cheap and simple to implement, yet not robust and accurate compared to the state-of-the-art deep learning algorithms, particularly when dealing with complex retinal data.

Below is a comparative analysis of some literature concerning the diagnosis of diabetic retinopathy using fundus retinal imaging systems. Various methodologies have been used including machine learning approach, deep learning, hybrid methods, and even conventional image processing. In the comparative analysis, various parameters like the methodology used, data set, strengths, limitations, and findings of the study have been taken into consideration. It is from such an analysis that the limitations and merits of current methodologies are identified paving way for motivation towards developing the proposed system.

Table 1 Literature Review Summary

Ref. No & Year	**Methodology**	**Dataset**	**Advantages**	**Drawbacks**	**Results**
1. (2025)	Review of ML (SVM, RF) and DL (CNN, transfer learning) techniques, comparison of different models, evaluation based on AUC, sensitivity and screening efficiency	Multiple datasets	Comprehensive overview of AI methods for DR screening	No experimental validation	Improved screening performance and reliable AI-based diagnosis

2. (2023)	Lightweight CNN (MobileNet-based) with optimized layers for reduced computation and faster inference	Fundus datasets	Fast, suitable for mobile/real-time use	Slightly lower accuracy	Fast detection with efficient real-time classification
3. (2023)	CNN with feature fusion (deep + handcrafted features) and focal loss to handle class imbalance	EyePACS / Kaggle DR	Handles class imbalance, improves stage classification	Complex model	Better stage classification and improved prediction accuracy
4. (2023)	Deep CNN models (ResNet, VGG) with preprocessing (normalization , augmentation) and feature extraction	Kaggle DR	High sensitivity, automated detection	Needs large labeled data	Demonstrates strong effectiveness of AI in DR screening
5. (2022)	Transfer learning using pretrained CNN (Inception) for predicting systemic health parameters from retinal images	UK Biobank	Predicts multiple health conditions	Low interpretability	Accurate detection with improved sensitivity and automation

6. (2024)	Hybrid CNN with lesion segmentation (microaneurys	DIARETDB1	Detects specific DR lesions, improves accuracy	High computational cost	Effective prediction of

	ms, exudates, hemorrhages) using image segmentation techniques				multiple health conditions from retinal images
7. (Year)	U-Net for segmentation + ResNet for classification to detect microaneurysms and early DR signs	DIARETDB1	Detects early DR signs effectively	Sensitive to noise	Reliable non-invasive screening with good classification performance
8. (2023)	Ensemble-based segmentation combining multiple models to improve lesion detection accuracy	Fundus datasets	Reduces false negatives	Time-consuming	Improved lesion detection and better clinical interpretability
9. (2025)	Hybrid CNN + Vision Mamba architecture for capturing both local and global features	Fundus datasets	Captures local + global features	Complex architecture	Early detection of DR signs with moderate accuracy

10. (2025)	Ensemble learning using multiple CNN models (ResNet + DenseNet) with voting mechanism	Kaggle DR	Robust and stable predictions	High training cost	High accuracy with enhanced feature extraction capability
11. (2022)	CNN combined with survival analysis to predict progression of DR over time	EyePACS (longitudinal)	Predicts DR progression	Needs long-term data	Stable and robust predictions with reduced classification errors
12. (2021)	Deep CNN for feature extraction from non-DR images to detect early diabetic patterns	EyePACS / Kaggle DR	Early detection without visible DR	Generalization issues	Improved detection rate with reduced false negatives
13. (2022)	CNN-based binary classification model for detecting diabetes from retinal images	EyePACS	Non-invasive diabetes screening	Lower precision than lab tests	Fast and efficient classification suitable for real-time use

14. (2025)	Image preprocessing (CLAHE, filtering) followed by CNN classification	Fundus dataset	Improves image quality & performance	Extra preprocessing needed	Improved detection performance using enhanced image preprocessing
15. (2026)	Traditional image processing techniques (thresholding, segmentation, feature extraction)	Small datasets	Simple and low cost	Low accuracy	Basic detection achieved with simple implementation

2.2 Drawbacks of Existing Approaches

In recent years, the number of studies focused on implementing the techniques of artificial intelligence and deep learning to detect diabetic retinopathy using retinal fundus photographs is high. These techniques have shown good advances in correctness and automation. Popular models used to analyze retinal images include convolutional neural networks, hybrid deep learning models as well as ensemble models. These systems can be used to minimize manual work and aid doctors with diagnosis. Nevertheless, despite all these advances, there are still a number of significant constraints, which impact their practical application and clinical utility. A significant weakness that has been seen in most of the current methods is binary classification. The output of such systems can be simply yes or no prediction of the presence or absence of diabetic retinopathy.

The other significant disadvantage is that despite the ability of many of these models to identify the retinal characteristics like microaneurysms, hemorrhages, and exudates, they are frequently employed to identify these characteristics. Still, doctors should

conduct further tests to comprehend the condition of the patient. This introduces a discrepancy between visual and clinical diagnosis. The high reliance on big and well-labeled data sets is another significant constraint. Deep learning models require a considerable set of retinal images that are annotated to train them. But in the medical profession it is not an easy and cheap task to gather such data since it must be annotated by an expert. Absence of interpretability is also an issue of concern. The majority of deep learning models are black-box systems with the internal working being not intuitive. It is a forecast as provided by the model but it fails to indicate the manner in which the decision was arrived at. This would pose a problem of trust in the field of healthcare where physicians would want to have systems that give clear explanations. The other difficulty is the generalization. The models are usually trained on a particular set of data and their efficiency is likely to reduce when applied to new data. The quality of images, types of cameras and patients may be different and influence the results. The other limitation is linked with the computational complexity. The majority of the advanced models are highly powerful in their calculations, possess a large memory and require much time to train. In the majority of health-care facilities, especially in rural

or low-resource areas, such infrastructure is absent. Such systems thus cannot be easily implemented in areas where they are most required. The other acute issue is that most of the existing approaches are mostly focused on improving accuracy as the main measure of performance. The other considerations are however equally considered in the medical applications such as sensitivity, specificity and reliability. Even a high accuracy system can fail in critical situations when it is not able to detect some specific conditions. Moreover, not every study includes proper clinical validation which reduces the trust to the systems. One more significant finding is that most of the current systems fail to take into account uncertainty during their predictions. With a medical model, one would desire to know the degree of certainty of the model and its output. However, most of the techniques merely provide a final prediction with no indication on whether the output is credible or not.

This may result in erroneous decisions when there is an uncertain forecast of the model. A proper system should be in a position to detect such cases after which the user should be warned in case of low confidence. Moreover, clinical knowledge is not integrated in these models to a large extent. Most of the deep learning models are not informed by the medical guidelines and domain knowledge and only image data is used. Consequently, the predictions might not necessarily be in line with clinical knowledge. Relations, e.g. the correlation of retinal damage and blood glucose levels can be added to the system to make it more useful. The second problem is the lack of continuous monitoring and follow-up analysis. The existing models are inclined to provide only one image prediction. They are not able to observe the long-term changes of the state of the patient. In medical practice, it is very important to monitor the progress of the disease in order to treat it successfully. Some systems also do not have well organized outputs that can be easily deciphered in the clinical setup. They are more likely to make raw predictions, not giving clear summaries, which may not be directly applicable in the decision-making process.

They tend to be concerned with only detection and fail to think about severity analysis, clinical interpretation, and real-life deployment issues. It is evident that a system that integrates different processes and provides valuable results to be used by medical professionals is needed. As such, this suggested approach will solve all the problems that are posed by current approaches because it provides an integrated approach in addressing this issue. In addition to recognizing the presence of diabetic retinopathy in individuals, the process also evaluates the intensity of the lesions as

well as measuring

the amount of blood glucose concentration in the body of the person in question. This is done through the initial process of segmentation for the detection of the important characteristics of the retina before conducting classification of the severity of the lesions. The proposed algorithm makes one step ahead in correlating these results to the health status of the individual in terms of the blood glucose and HbA1C level.

CHAPTER 3

PROPOSED METHOD

3.1 Architecture diagram and proposed method

An architectural diagram is a graphical representation of the system showing how various components are organized and their interactions with each other. An architectural diagram provides a very comprehensive explanation of the whole system, showing how data flows within the system, and how processes occur within the system. Using an architectural diagram helps one to have a better understanding of how the system operates from end to end, knowing what should follow when certain inputs are given leading to a certain output. In other words, using an architectural diagram makes complicated systems easy to understand through various simple steps.

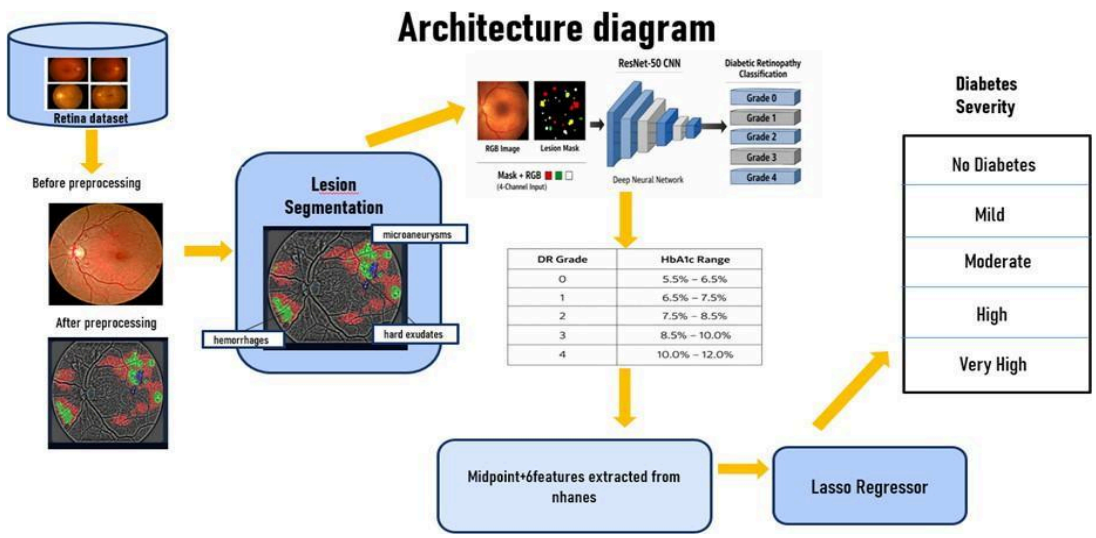


Figure 2 Architecture Diagram of Proposed Work

From Figure 2 we can see the proposed system architecture which is a complete end-to-end pipeline to predict the severity of diabetes and estimate the levels of blood glucose through the utilization of retinal fundus images. The system is structured and modular with each stage having a particular task and its output is passed to the next stage. It starts with the input of the fundus retinal images in a series of images. Preprocessing is the initial process since raw images might have noise, uneven lighting, and undesired boundaries. This is a stage in which the black borders are removed leaving only the useful retinal area. The brightness of the image is evened by illumination correction, and fine details, like blood vessels and lesions are enhanced by the contrast enhancement with the CLAHE. The image is also resized to a standard

image to be effectively processed by deep learning models. This will purify the input that will be uniform and fit to continue with the analysis.

Once the preprocessing is done, the system enters into lesion segmentation phase. At this stage, pertinent retinal anomalies are detected using a deep learning model. The model identifies lesions such as microaneurysms, hemorrhages and hard exudates that are the primary symptoms of diabetic retinopathy. The result is a segmentation mask which points out the affected areas in the retina. This mask provides specific details on the location of the damage and type of damage and this is critical to improve accuracy of the next step. The second one is the classification in which the severity of diabetic retinopathy is determined. The processed image has the lesion mask placed on top of it to create a four-channel input. The sum of these inputs enables the model to operate on the visual features as well as the lesion specific information. This information is inputted into a deep learning model to estimate the DR grade, Grade 0 to Grade 4. This conversion converts the analysis in the shape of images into a considerable level of severity.

The DR grade is then extrapolated to an HbA1c range using clinical knowledge after receiving the DR grade. Every grade is associated with a particular range of HbA1c which indicates the average blood sugar level of the patient over a certain period of time. The middle of this range is taken as an approximate of HbA1c. This is an intermediate step between the output of the deep learning model and clinical interpretation. The next step in feature engineering is undertaken. The system creates more features based on the HbA1C value and DR grade. These features can help to obtain correlations between glucose levels and severity of the disease. The feature vector generated is then fed into the prediction model. Finally, the system uses a regression model to predict the amount of fasting blood glucose. The model used is Lasso Regression as it minimizes overfitting and uses significant features. The model is fed with the input features to give the total glucose value in mg/dl. This is then displayed to the user as an approximation of the level of blood glucose and the level of diabetes which is none, mild, moderate, high or very high. This makes the system useful both to the clinicians and patients to know and monitor their status. The architecture in general provides a general and effective non-invasive diabetes predictive solution. The system also offers the image preprocessing and lesion detection, classification and regression to guarantee the most suitable results.

3.2 Modules in Proposed Work

The system splits everything into separate modules. Each one handles its part of the process, so you always know what's happening and when. Building it's a breeze, too, and if something breaks, you're not stuck sorting out a mess—you just fix the problem module. These modules pass the retinal image along like links in a chain, each transforming it a bit, until you end up with a blood glucose prediction.

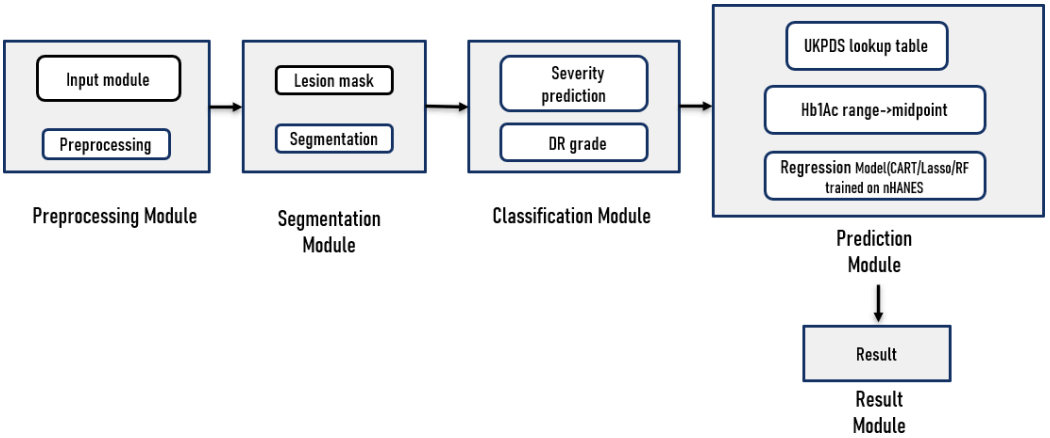


Figure 3 Module Connectivity Diagram

The proposed algorithm "Diabetes Severity Prediction from Retinal Fundus Images" (GlucoVision AI) is expected to deliver an integrated intelligent system that includes not only image processing but also machine learning and clinical predictions. The input to the developed algorithm is supposed to be a retinal fundus image, while its output is a patient's blood glucose level. Owing to the complexity of the task in question and proper performance of all the necessary actions, the design of the algorithm is going to involve several modules collaborating with each other and performing particular functions. As shown in Figure 3, the system architecture, it can be said that its tasks can be listed as follows: image pre-processing, lesion detection, severity prediction, and clinical prediction.

3.2.1 Input and Preprocessing Module.

Input and Preprocessing Module is the base of the whole system. It does the task of preparing the raw retinal fundus image in an efficient manner which can be subsequently processed by the deep learning models to give the desired output. In real world conditions, retinal images are frequently subjected to noise, uneven lighting,

low

contrast, and superfluous background areas. Unless these problems are resolved, it could negatively impact the system performance. The module starts with an input of a retinal fundus image. The first step of preprocessing is to remove the black borders that tend to surround fundus images. These are non-informative boundaries that may bring unwarranted noise during the learning process. This removes these areas thereby concentrating on the useful retinal area in the system. This is followed by illumination correction by use of the Ben Graham method. The step removes the differences in brightness across the picture as well as enhancing the overall visibility light imbalances. The problem of light imbalances in general in medical imaging will be addressed by light correction to ensure that the significant features will not be lost behind the light imbalance. This is then contrasted by enhancement using the CLAHE on the L-channel of the LAB color space. The technique boosts a local contrast without excessively amplifying noise. As a result, tiny lesions and other subtle details of the retina become more evident.

Lastly, the picture is down sampled to a constant resolution of 512 × 512 pixels. This provides consistency in the size of the input of all samples and also makes the data capable of being processed by the deep learning models. The preprocessing module standardizes the input image, cleans it and makes it fit to continue with the analysis.

3.2.2 Segmentation Module

Segmentation Module identifies and localizes significant retinal characteristics, called lesions, which are good indicators of diabetic retinopathy. This module is vital in deriving meaningful information of the image on the pixel level. Segmentation is done using a U-Net architecture that uses an EfficientNet-B4 encoder. The encoder is the one that extracts high-level contents of the image and the decoder is the one that reconstructs the details of the spatial contents to build the segmentation map. This is a combination that allows the model to store both the fine-grained and global information. The model is trained on the FGADR data; this data is composed of annotated retinal images with lesions detected. The microaneurysms, hemorrhages, hard exudates, soft exudates, neovascularization and IRMA are the identified types of lesions in the segmentation process. The different degrees and stages of diabetic retinopathy are such lesions.

One of the biggest issues of segmentation is the class imbalance since certain types of lesions are not as prevalent as others. This is restricted by combining Dice loss and Focal loss. The dice loss is applied in cases where one is interested in an overlap

between the predicted and the actual regions whereas Focal loss is applied where one would wish to give more attention to challenging or less frequent samples. This improves the lesion detection ability of the model in detecting any form of lesions. The product of this module is a multi-class lesion mask, in which the pixel falls into a certain category. This mask provides a detailed view of the retinal damages and could be a major contribution to the classification aspect.

3.2.3 Classification Module

The Classification Module classifies the severity of the diabetic retinopathy based on the original retina image and the lesion data available to it obtained by the segmentation module. This stage enhances the process of decision-making, since it deciphers visual properties and detects abnormalities. The system makes four channels of input as opposed to three RGB channels by overlaying the three RGB channels with the segmentation mask to the RGB image. This allows the model to focus more on regions where the lesions are and improve the accuracy of classification. This is done using a ResNet-50 architecture. This deep convolutional neural network can do this, so it is capable of extracting the complex features and learning the hierarchy representations of the image. The input categories have five classifications indicating the degree of severity with the first being no diabetic retinopathy to the extreme. Weighted random sampling is used during training to deal with the disparity in the distribution of datasets. This will ensure that the underrepresented classes are well attended to. The model is trained by the Cross Entropy loss, which is suitable in the case of multi-class problems of classification. One more characteristic of the given module is the presence of a confidence threshold. Uncertain predictions are indicated as low confidence, and can be revisited. This enhances reliability in the system and minimizes the chances of false predictions being blindly accepted. The outcome of this module is the forecasted grade of diabetic retinopathy, the extent of retinal damage.

3.2.4 Prediction Module

Prediction Module is where the prediction of the degree of severity becomes medically useful. This is the point where the significance of our work in application to medical needs becomes clear. The first step in our model involves mapping the predicted level of DR severity to a specific HbA1c level using a lookup table, as shown in Table 2 defined by UKPDS guidelines. In general, HbA1c refers to the average glucose level in blood over a certain period of time. In our case, we link

retinal damages with the metabolic state by defining how HbA1c correlates with DR severity. Based on the

estimated range of HbA1c, we calculate its mean which would give us an estimation of HbA1c level.

Table 2 UKPDS Lookup

DR Grade	HbA1c Range
0	5.5%-6.5%
1	6.5%-7.5%
2	7.5%-8.5%
3	8.5%-9.5%
4	10.0%-12.0%

We then use that particular level to calculate other features such as squaring it and forming interaction features of both. These features are then provided to the different regression models such as Lasso Regression, Decision Tree (CART), Random Forest and Baseline Formula. Models are trained based on the NHANES data set which provides medically accurate correlation between blood glucose level and HbA1c level. Metrics that measure model performances include MAE, RMSE, and R^2. Based on the above observations, Lasso Regression would be selected as the best choice for the model because it minimizes error without overfitting. The final result from this module analysis will be the estimated value for fasting blood glucose concentration in mg/dL.

3.2.5 Result and Visualization Module

This module involves presenting output generated by the system in a format that can be easily understood by the user. This result module ensures that not only the correctness of the output is guaranteed but also its clarity by the user. The output is composed of the estimated blood glucose levels, DR severity, and a confidence score. Furthermore, the system provides visuals like overlays of lesions on retinal images. These visualization outputs help the user to identify the part of the image responsible for making predictions through the model. The output process is facilitated through an interactive output screen.

3.3 Analysis and Design through UML

Design and analysis using UML involve designing a system in an analysis and design approach by using UML diagrams. UML is a language used to design a system with features such as interactions, relationships, and others. In a software development

process, it is mandatory to conduct an analysis of the system and then its design prior to implementing the system. The analysis process aims at understanding the system functionality, whereas the design process seeks to understand the system implementation. UML diagrams are used to plan and design the system systematically in these two processes. As far as the proposed system is concerned, the application of UML lies in representing how the retinal images will be processed in different stages, such as preprocessing, segmentation, classification, and prediction stages. UML can also be used to demonstrate the interfaces of the users and the interaction among the system's internal modules. It should be noted that UML depends on diagrams to represent the systems; therefore, UML diagrams are significant in designing a system. These diagrams are useful because they provide a visual representation of the systems, which makes explanations easier as compared to textual explanations. Besides, diagrams can highlight errors in the system design process.

Different kinds of UML diagrams can be used for designing systems.

1. Use Case Diagram
2. Activity Diagram
3. Sequence Diagram
4. Class Diagram
5. Data Flow Diagram
6. Control Flow Diagram

The use of UML for our project will aid us in properly structuring our system architecture, facilitate better communication.

3.3.1 Use Case Diagram

A Use Case Diagram basically shows you what interacting with a system looks like in real life. Say you upload a retinal fundus image and click submit. After that, you don’t need to do anything else,the system handles everything from analysis to giving you a predicted blood glucose level. Your job ends with the upload. The diagram maps out all the possible things users can do and what kind of results they’ll get, letting you see the whole process at a glance. You find out exactly where you come in, what steps you need to take, and how the system reacts. It’s a straightforward way to understand the entire experience from start to finish.

The use case diagram in Figure 4 shows the interaction of the external users with the proposed system, Diabetes Severity Prediction using Retinal Fundus Images. There are two key participants in this project namely the Patient and the Doctor (Clinician).

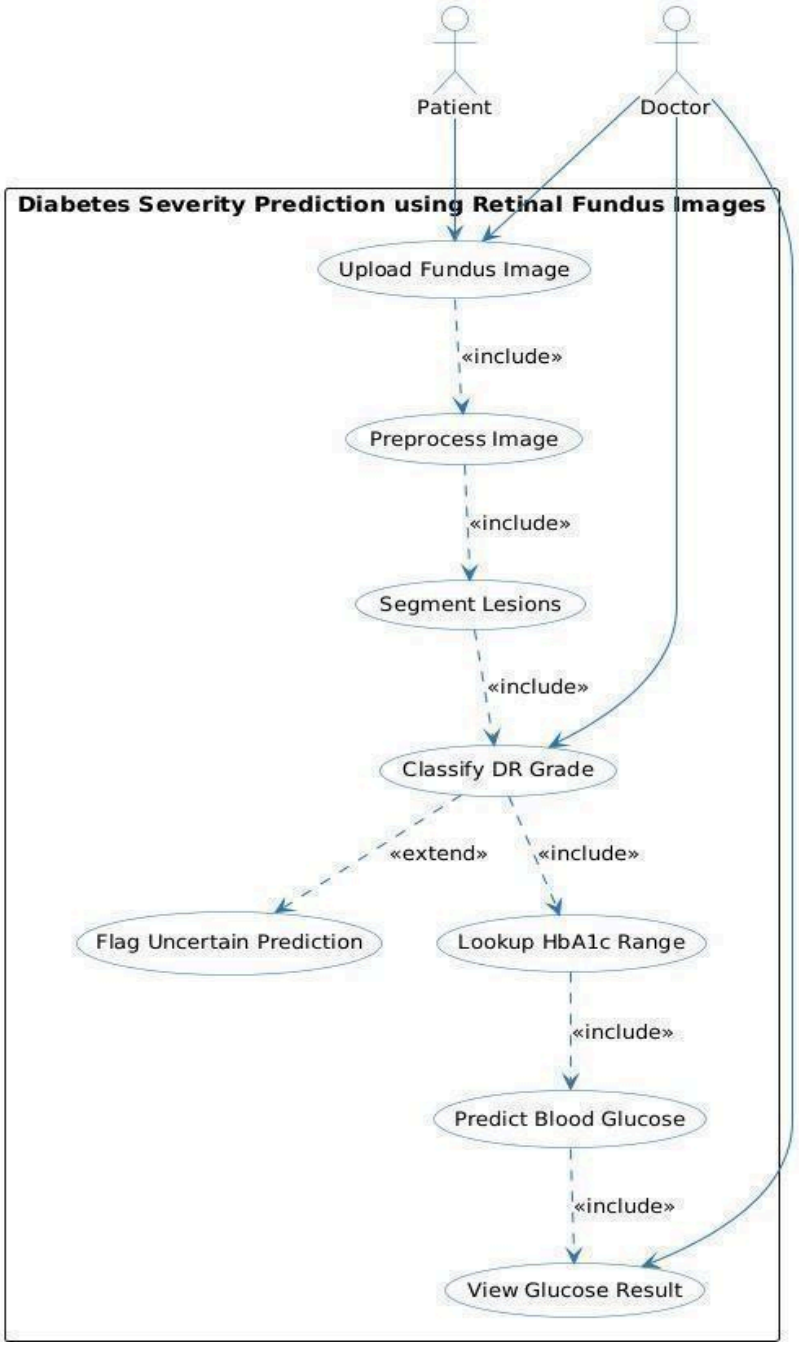


Figure 4 Use Case Diagram

This starts by the patient or the doctor uploading an image of the retinal fundus into the system. The Upload Fundus Image use case directly relates the two actors since both are capable of starting the process. Once the image has been uploaded, the system automatically performs additional tasks without any additional manual aid. Image preprocessing is the next step. This use case is added to the upload procedure i.e. it is automatically triggered when an image is provided. Here, the system maximizes the image quality by removing the unwanted borders, adjusting the lighting and improving the contrast. The user is not directly involved with this step

but it is an essential part of

the inner workings of the system.

The system also after preprocessing would go through the segmentation stage. This too is represented as a second use case that is included meaning that it is an obligatory step in the pipeline. In this stage, the system recognizes the retinal features that are crucial such as microaneurysms, hemorrhages, and exudates. These features are crucial in the detection of the presence and the extent of diabetic retinopathy. The second important use case is referred to as Classify DR Grade and the system analyses the processed image and provides the level of severity of diabetic retinopathy. This is an important step to the system as it will directly affect the end prediction. It is an auto-implemented classification not requiring a user. Two are the potential routes beginning with the stage of classification. When the system realizes that it is not sure about its prediction, it invokes the Flag Uncertain Prediction use case. This can be represented as an extended relationship, that is, it can only happen under some conditions. This factor enhances the accuracy of the system since it indicates when the forecast may not be absolutely valid. In the meantime, the system continues with the use case of the Lookup HbA1c Range that is represented as an «include» relationship of the classification stage. This means that it is never done before being classified. The predicted DR grade is mapped to a specific range of HbA1c values in this step via clinical knowledge. This helps to fill the gap between image analysis and medical interpretation. The next system of the workflow is the use case of Predict Blood Glucose. The HbA1c and DR grade would be applied in this case to predict the amount of blood glucose of the patient using regression models. This process transforms the output of a classification output into clinically meaningful value. Finally, the View Glucose Result use case is to allow the user to see the result. This outcome may be achieved by the patient and the doctor. The result is usually the estimated level of blood glucose, the grade of the DR severity, and other information such as the level of confidence or the regions of the lesions highlighted. The doctor can use this information to diagnose and make decisions and the patient to be aware and monitor.

Overall, the use case diagram is eloquently used to illustrate how the system communicates with the users. It highlights the reality that the majority of the processing steps are automated in the system and the ease of provision of input and observation of the results. The fact that we use «include» relationships to model mandatory steps in the workflow and that we use the relationship «extend» to model

conditional behavior,

e.g. to deal with uncertain predictions. The ordered representation helps to understand

the functioning of the system and ensures that all the required operations are specified properly.

3.3.2 Class Diagram

The system structure is outlined clearly and in a systematic manner in the Class Diagram. It illustrates how the different aspects of the system are divided into classes, their attributes, methods and how they interact with each other. Each of the classes is a specific functionality of the system and the combination of all the classes becomes the whole of the pipeline. This chart can help in understanding the internal architecture of the system and how the data flows in and out of the different modules. The system contains the most important classes which are the image processing class, the segmentation model class, the classification model class and the prediction module. These dominating classes have other supporting classes which ease data and result management. These classes retain the intermediate findings such as lesion masks and classification findings and they enable easy cross-communication between modules. Associations are used to depict the relationships among the classes with one class utilizing or relying on another to carry out its work. In general, the class diagram gives a clear picture of the interior design of the system. It helps to learn the composition of different components, and how they interact to the ultimate goal of determining blood glucose levels based on retinal images.

The class diagram from Figure 5 shows the internal structure of the proposed system by indicating the different classes, their attributes, functions and the relationships between the different classes. It explains the logical organization of the way the system is assembled, and the data flow between the different parts of the system. In this project, the entire pipeline is modeled using the class diagram that starts with the image input and ends with blood glucose prediction. There is a top most system called Inference Pipeline. This course is the main organizer which connects all the other modules.

It includes references to various elements, including the preprocessor, segmentation model, classification model, clinical look up, feature engineering module, and prediction model. The main purpose of such a class is to execute the whole workflow by calling each of the modules in order to get an image as input. The procedure starts with the Fundus Image class, which is the input retinal image.

Basic information like image ID and file path is stored in this class. It has also a procedure of loading the image into the system. This course is the initial step in data

processing. The picture is subsequently sent to Preprocessor class.

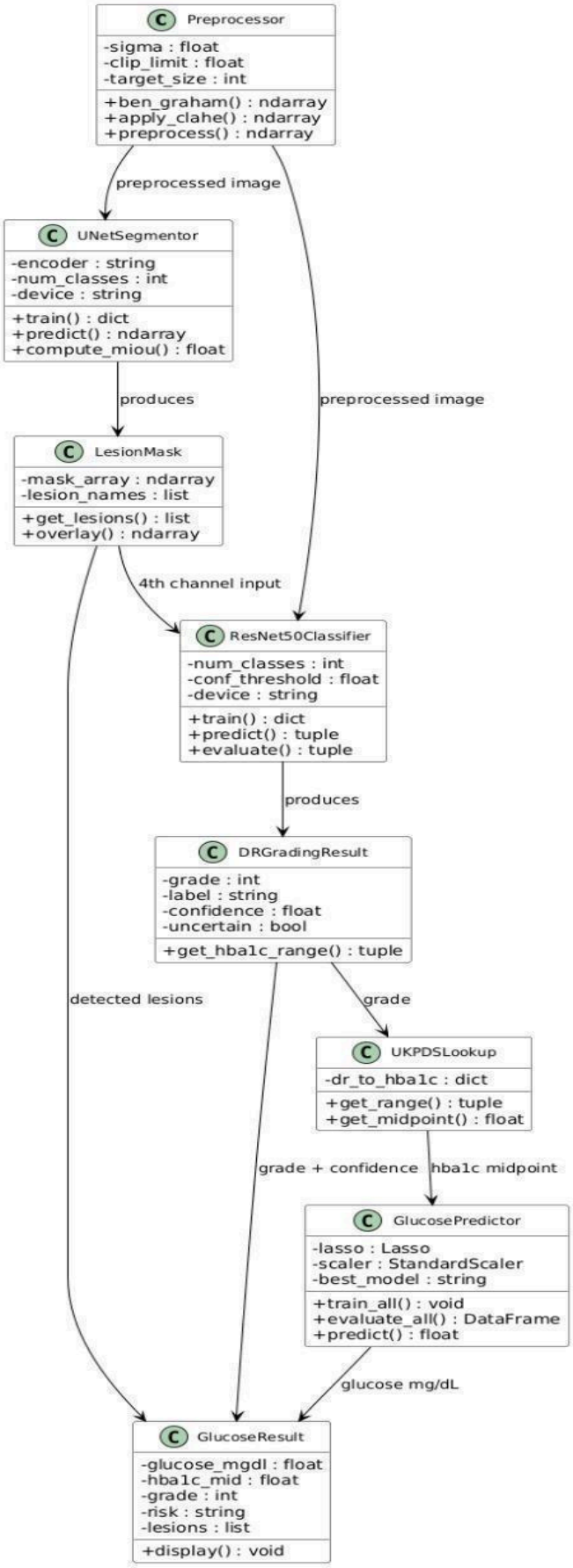


Figure 5 Class Diagram

This class is involved with improving the input image. It has parameters of sigma value, clip limit and target image size which regulate preprocessing activities. The techniques in this course are to eliminate black borders, use illumination correction and increase contrast with CLAHE. The output of this kind is a standardized image which has been denoised and could be further processed. The preprocessed image is then provided in the U Net Segmentation class. This class does lesion detection based on a deep learning segmentation model. It has such elements as the kind of encoder, the amount of classes and the calculator. The most important activities of this class are to train the model and predict lesion masks of input images. The result is a segmentation mask which emphasizes disparate lesion areas. The outcome of the segmentation is saved in the Lesion Mask class. In this class, the mask array is stored and a list of identified lesion types. It also offers how to get detected lesions and superimpose them to the original image. The class acts as a bridge between segmentation and classification of well-organized information about lesions. The system then decides the level of diabetic retinopathy with the use of the ResNet50Classifier class. The number of classes, confidence threshold and the computational device are attributes in this class. The input to the classifier is a combination of the original image and lesion mask and the output is a prediction. The findings are placed in the DR Grading Result class whose contents are DR grade, label and confidence score and uncertainty flag.

The projected DR grade is then sent over to the UKPDS Lookup class. In this course a mapping of DR grades with the HbA1c ranges of clinical data is made. It provides means of getting the interval of HbA1c and calculating the mean. This step will help translate the outcome of the classification into a clinically significant parameter. This system is then transferred to the Feature Engineer class which creates more features depending on HbA1c and DR grade. These features consist of transformed and interaction values which can be used to enhance the accuracy of prediction. The result is a feature vector which describes the condition of a patient in a formal way.

Lastly, the feature vector is sent to the Glucose Predictor class. Some of the regression models in this class are lasso, decision tree and random forest. It picks the most appropriate model according to performance and forecasts the end value of blood glucose. The result is stored in the Glucose Result class containing the glucose level, HbA1c level, DR grade, risk level and lesions identified. It also provides a method to display the result. All these classes are connected to each other with the help of

relationships that are required by a class and are provided by the other class. This is all

organized by the Inference Pipeline class to make the system a whole. In general, the class diagram gives a vivid idea of the internal structure of the system. It demonstrates the interaction of various modules, data transformation at each point and generation of the final output. This helps in designing the system and all the components are properly linked and mounted.

3.3.3 Sequence Diagram

It would be correct to describe the Sequence Diagram as a method through which interactions of different components in the system are illustrated. In particular, the Sequence Diagram focuses on demonstrating the sequence of operations conducted and how data flow from one component to another. Understanding of dynamic interaction in the system is possible using the Sequence Diagram since it illustrates how different components respond to input information and how they interact among themselves during processing. Thus, this method provides a possibility to illustrate the entire process of conducting operations. Moreover, it allows illustrating control flow between different components in the process of operation performance. Thus, use of the sequence diagram makes it possible to get an idea of relationships between components, sequence of actions, and missing components.

The sequence diagram shown in Figure 6 illustrates the process of interaction between various entities of the Gluco Vision AI system as they respond to the retinal fundus image as time goes by to come up with the final prediction of blood glucose. It shows the exact sequence of starting of all the modules and flow of data through the system step by step. This starts as a user (a clinician or patient) uploads a retinal fundus image via the system interface.

The main inference pipeline gets this image and serves as the controller and oversees the running of all modules. The input image is first fed into the preprocessing module by the pipeline. The system enhances the quality of the image during the preprocessing stage prior to analysis. The module eliminates black edges on the fundus image, uses a method by Ben Graham to correct illumination to normalize the brightness and uses CLAHE to boost the contrast of the L-channel of an LAB color space. The picture is then rescaled to a fixed size of 512 x 512 pixels. The enhanced image is then sent back to the pipeline after preprocessing and then to the segmentation module.

The segmentation module relies on the U-Net model, which contains an EfficientNet-B4 encoder, which examines the image on a pixel level. It determines the

different types of retinal lesions such as microaneurysms, hemorrhages, hard and soft exudates,

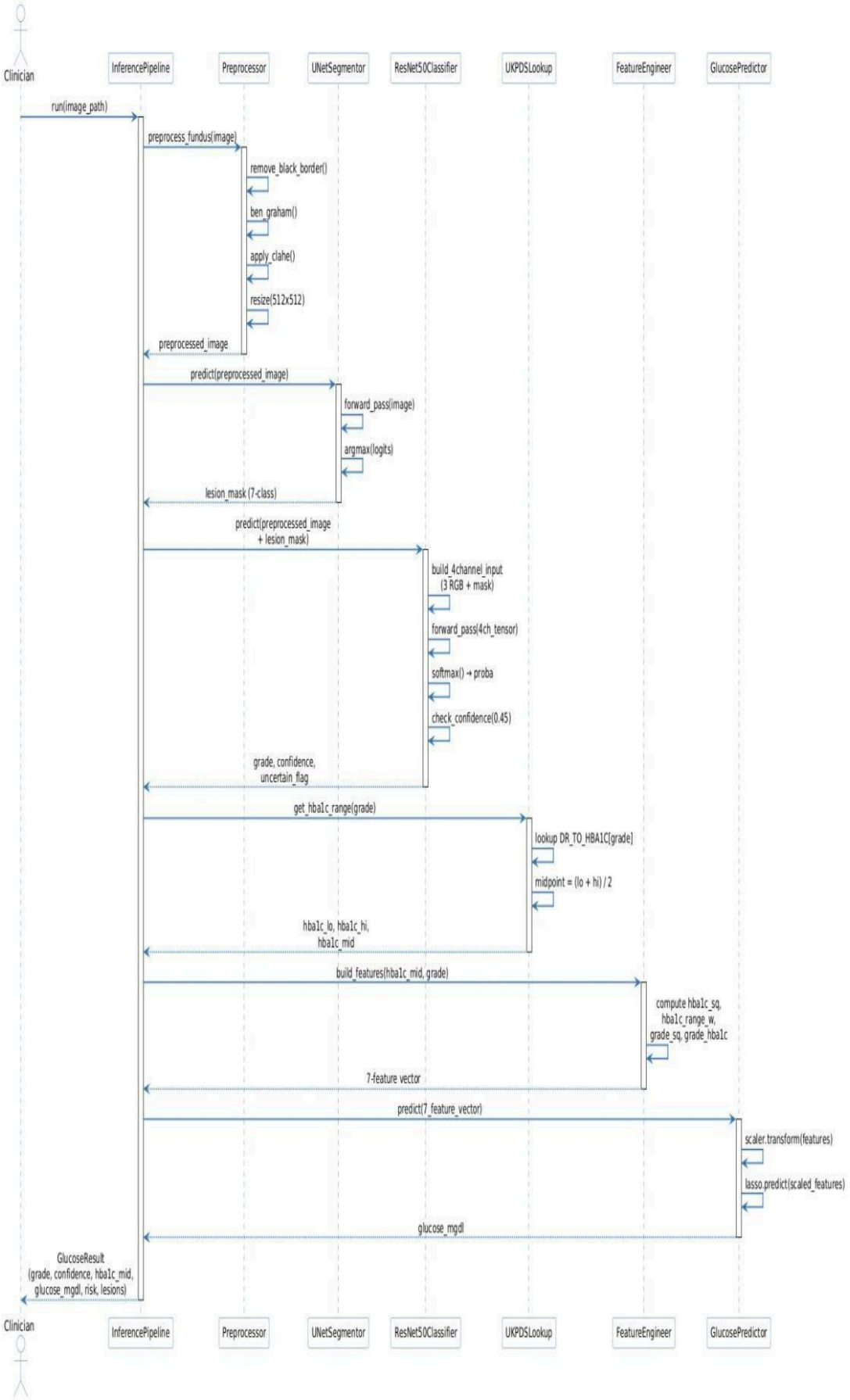


Figure 6 Sequence Diagram

neovascularization and IRMA. This analysis results in the generation of a multi-class lesion mask that determines the lesions in the retina. This mask is re-sent to the pipeline. The original preprocessed image and the generated lesion mask are then merged to generate a four channel input by the pipeline. This combined information is sent to the classification module. The classification module is based on the ResNet-50 model, which is used to determine the severity of diabetic retinopathy. It evaluates the visual characteristics and lesion details to give a grade (0 to 4). The model also produces the score of confidence and the estimated grade. After classifying the pipeline, it looks at the confidence score. When the confidence is below the predefined level, the system reports the outcome as uncertain, implying that more investigation of the prediction may be required. Where there is an acceptable confidence, the grade of DR forecasted is sent to the clinical mapping module.

The grade of DR is converted into the range of HbA1c values at the clinical mapping step with references to the UKPDS clinical standards in a look-up table. This range is then calculated to give the midpoint of this range which is an approximated HbA1c level that represents long-term levels of blood glucose. The DR grade, along with this value of HbA1c, is sent to the feature engineering module. This module generates other characteristics, squared terms and the value of interaction, which is the relationship between the severity of the disease and the level of glucose. These features make up a structured input to the predictive model. These characteristics are then inputted into the prediction module whereby they are computed by a number of regression models which comprise of Lasso Regression, Decision Tree and random Forest. These models are trained on real world medical data, namely NHANES dataset, relating HbA1c to blood glucose levels. The best of these models is selected based on the performance measures of MAE and RMSE. The chosen model subsequently forecasts the ultimate level of fasting blood glucose in mg/dL.

Lastly, the resultant glucose value is sent back to the inference pipeline and shown to the user via the system interface. The system also shows the DR severity grade, confidence score and visual representation of lesions detected besides the glucose level. All this is to ensure that the system does not only process the image, but also provides some meaningful and interpretable medical output.

3.3.4 Activity Diagram

Activity Diagrams are a visual depiction of the process flow, showing all the processes involved to carry out certain actions to transform the input into output

products.

Activity Diagrams are made to illustrate a specific set of actions carried out in a sequence in the process flow. This can be used to analyze the entire process flow. In this scenario, the activity diagram illustrates the entire process flow of the Gluco Vision AI app, starting from the very beginning till the end. In the first step, an image of the retinal fundus is required to be submitted by the user to the software as input data. For this, users have to upload the image file within the software interface. To conclude, the activity diagram is a great tool for showing the processes involved in the system. This is because the activity diagram shows the process performed by the system, the decision taken and the data flows involved in each process.

Activity Diagram in Figure 7 is a workflow overview of the proposed system and is general as it illustrates how different activities are carried out in a specific order where input is the starting point and output is the end point. It illustrates the processes of control, data circulation and decision making that are involved in the production of the final end result. This is activated when a retinal fundus image is uploaded in the system by the patient or doctor. Once the image is received it is processed to the preprocessing stage. At this step, techniques like black border removal, Ben Graham illumination correction, CLAHE to increase contrast and resizing the image to a common size of 512*512 are used to clean and enhance the image. These are the steps that make sure that the image is normalized and can be used in further processing. After the preprocessing, the workflow passes through the segmentation stage.

In this instance, the forward pass is implemented with a deep learning model (U-Net) to make a prediction of a multi-class lesion mask. The system identifies different types of lesions that consist of microaneurysms, hemorrhages, hard exudates, soft exudates, neovascularization and IRMA. A lesion mask, which points on the abnormal areas in the retina is the output of this stage. The second stage is the classification stage. The system builds a 4-channel input to combine the image of the lesion produced with the RGB image to form the lesion mask. The input is fed to a ResNet-50 model that does a forward pass and generates a DR grade and a confidence score with the help of a softmax function. The confidence value is then given a decision point. The uncertainty of a prediction of less than 0.45 means that the system is uncertain. Otherwise, there is a continuation of the workflow to the next level.

An HbA1c range is the predicted DR grade is then mapped to a range of HbA1c in a UKPDS table of clinical mapping in the clinical mapping phase. The calculated value of the midpoint in this range is taken as an estimated HbA1c value. After this, the

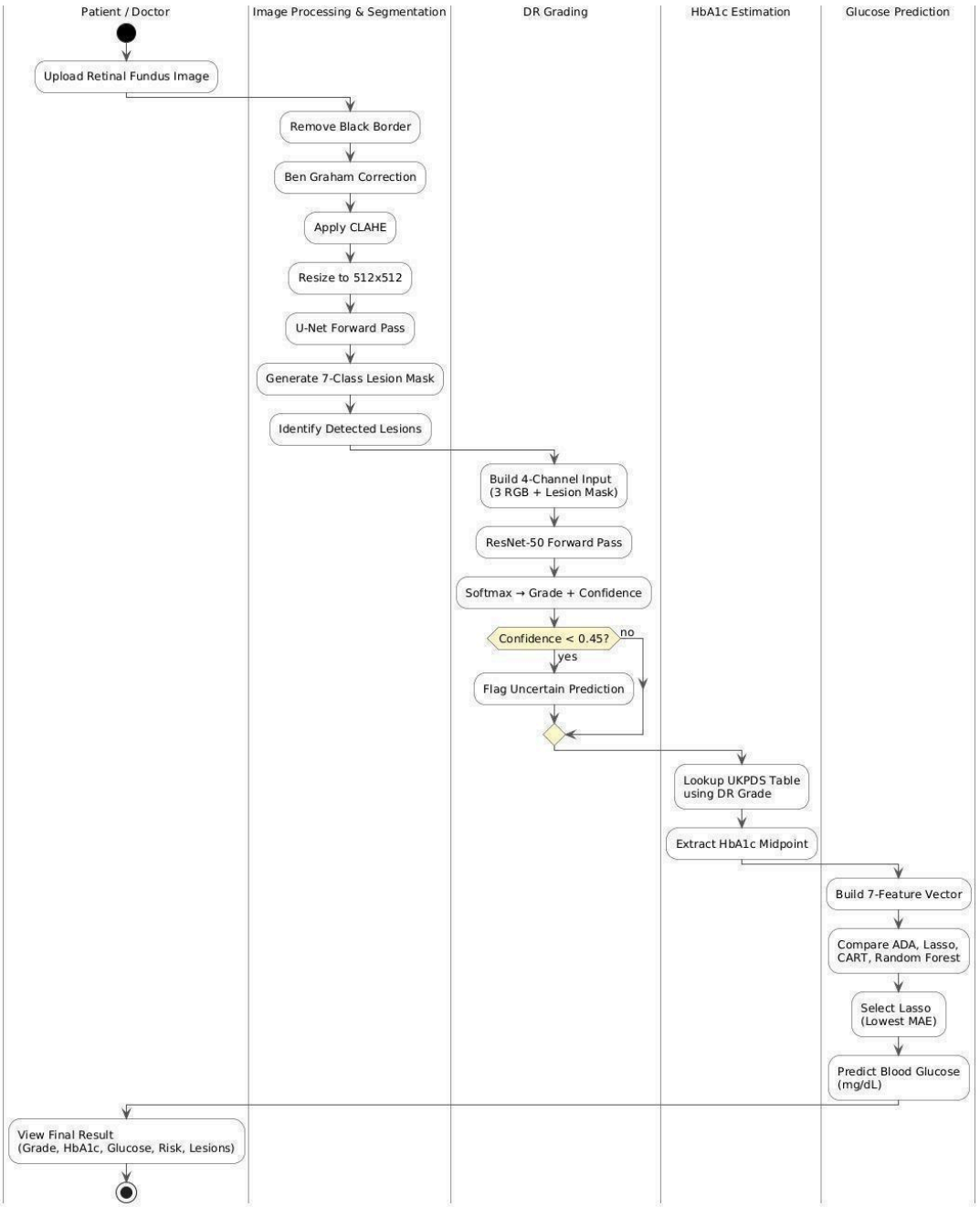


Figure 7 Activity Diagram

system carries out feature engineering. The HbA1c midpoint and DR grade are used to create a feature vector based on which the prediction is built in a structured way.

This is followed by a glucose prediction step. Many regression models such as ADA baseline, Lasso, CART and random forest are run and the best model (Lasso, based on the minimum MAE) is selected. The model is used to predict the ultimate blood glucose level in mg/dL. Lastly, the system shows the outcome, which consists of the DR grade, HbA1c level, the predicted value of blood glucose, risk level, and lesions detected. This fulfills the system workflow.

3.3.5 Data Flow Diagram

Data flow diagram is one form of a graphical tool that depicts how data flows through a certain system. Using DFD creates an easy way of visualizing the flow of data in the entire system. There are four basic components of drawing a DFD and they include: process, data store, external entity, and data flow. DFD uses simple diagrams that indicate how data flows through processes in a system. DFD focuses mainly on how data flow is used to achieve meaningful results. The main benefit of using the DFD is that it gives an overview of the operation of the process and transformation of data in the process. In the current project, the DFD describes the processing of the retinal image. The diagram shown in Figure 8 is an official pipeline where all the modules contribute to the conversion of a retinal fundus image into a clinically useful prediction. The system is designed in a module and step by step fashion with the output of one stage serving as an input of another to enable the data to flow seamlessly and enhance the interpretability. The retinal fundus image is captured by the user interface in Data Receive. This is to ensure that the input image is loaded and passed but is also checked. The retinal images are subject to size variations, variations in quality and lighting conditions and as such, the system must normalize the images and subsequently analyze them.

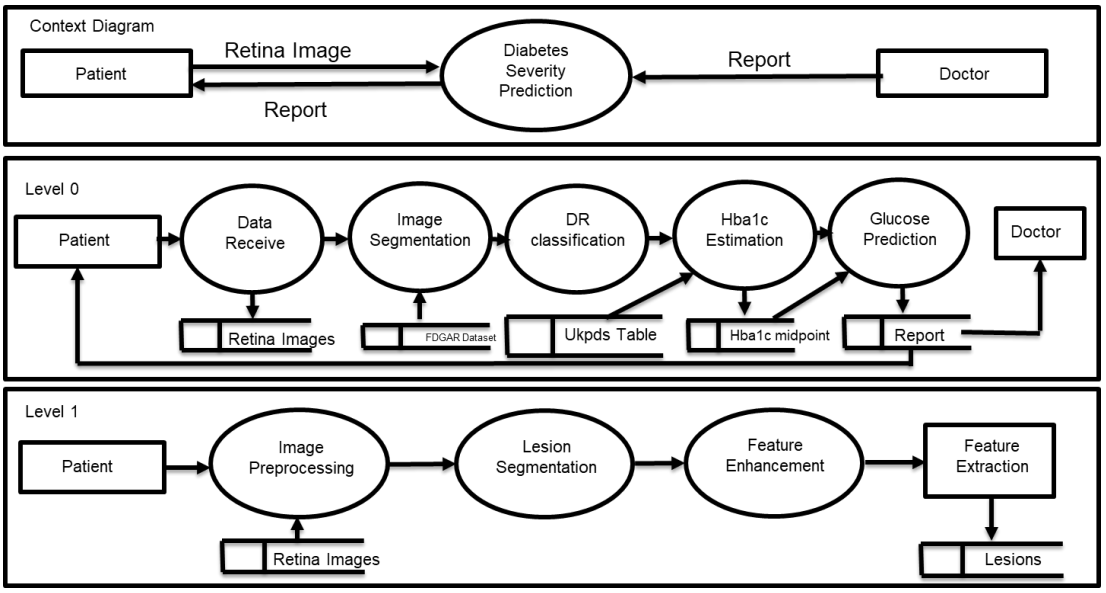


Figure 8 Data Flow Diagram

The next process is Image Preprocessing which is significant in improving the quality of data. Raw retinal images are typically defined by black edges, uneven lighting and noise. Filtered approaches, such as the removal of black borders, illumination repair (Ben Graham method) as well as contrast adjustment with CLAHE are applied. The

image size is also cut down to a standard size (e.g. 512x512) so that they are the same size across the dataset. This step enhances the salience of important structures like blood vessels, lesions and it becomes more convenient to obtain meaningful features by the deep learning models. The system may then proceed to Image Segmentation (after preprocessing) in which the retinal area is processed to determine the important anatomical structures. This is to reduce the irrelevant background data and to focus computation on areas of significant importance on the retina.

It is an intermediate between image refinement and lesion analysis. Lesion Segmentation, the next step, is more specific and medically significant. The classification of lesions on a pixel-level is done with the help of a deep learning model, such as U-Net. Some abnormalities detected by the system include microaneurysms, hemorrhages, hard exudates, soft exudates, neovascularization and IRMA. The outcome is a multi-class lesion mask that reveals the localization and the nature of the damage. This process is interpretable as it will give a visual demonstration of where the disease is located. As soon as lesions are detected, the system enters Feature Enhancement. In this step, the preprocessed RGB image will be multiplied with the lesion mask to form a multi-channel input (typically 4 channels: RGB + mask). This allows the model to store image features on a world scale and features on small lesions. It is also possible to obtain other engineered properties to improve the representation. This is a significant measure because it improves the learning capacity of the classification model. These enhanced features are used to perform DR Classification in the system. A deep learning model such as ResNet-50 is used to work with the multi-channel input to predict the severity of diabetic retinopathy. The outcome is the score of 0-4, which is the severity of the disease. The model also comes up with a confidence score that is used to inform how confident the prediction would be. This is a step that transforms patterns seen into a medical category order.

The second step is HbA1c Estimation, and the estimated DR grade is then translated into a clinically significant range of HbA1c based on a look-up table, informed by medical principles (e.g., UKPDS). HbA1c is a measure of the average blood sugar level. The mean of the mapped range is an approximation of the HbA1c. This is an intermediate between image and clinical analysis. Finally, estimated HbA1c and derived features are calculated by the system to forecast the amount of fasting blood glucose using regression models such as Lasso, Decision Tree, or Random Forest. The

best of these (e.g. the Lasso based on the lowest MAE) is selected to produce the final

value in mg/dL. It is then possible to give the results to the user as DR grade, confidence level and lesion visualization, making it informative, and interpretable.

3.3.6 Control Flow Diagram

Control flow diagram (CFD) is the name of the flow chart used to show the flow order of execution of the processes in the system. CFD contains all the flow control processes from one process to another and all the decisions to be made at different levels and how these conditions dictate the path. In contrast, while data flow diagrams concentrate only on flow of data, flow control and the order of executing the processes are concentrated in control flow diagrams. Considering the case for the current system, the control flow diagram shows the process of analysis of retinal fundus image to come up with blood glucose prediction results. From this flow chart, we can see that the first step is image input and validation followed by image segmentation and lesion detection processes. Depending on the decisions at hand, the flow will be decided until finally, blood glucose prediction results will be gotten.

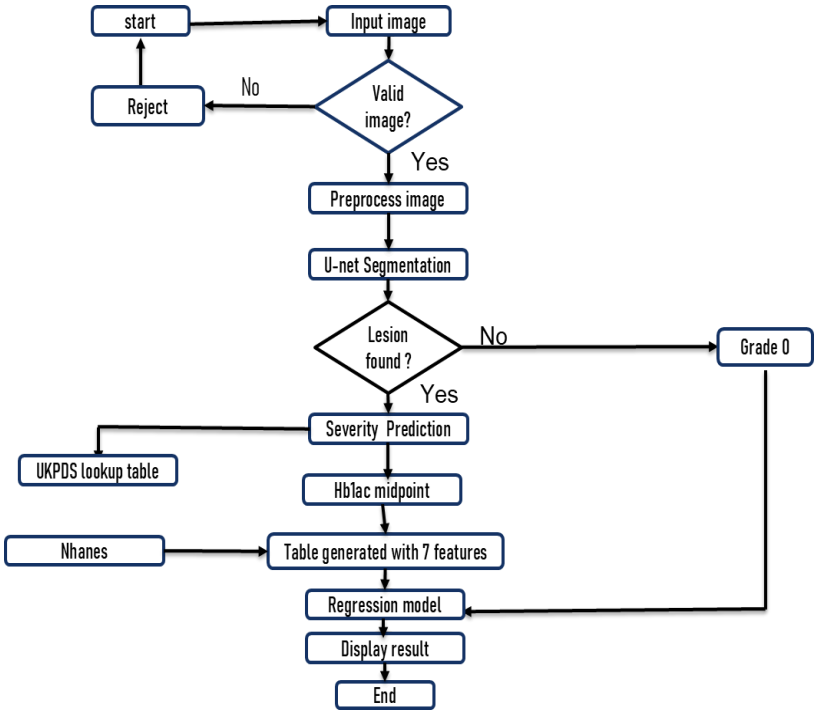


Figure 9 Control Flow Diagram

Figure 9, the Control Flow Diagram, provides an excellent example of the control flow between the different stages of the proposed system. It not only shows the sequence of execution, but also shows where decisions are made where the system behavior differs with the circumstances. This helps in understanding how the system ensures that there is correctness and reliability in all the steps. This begins with the

start node followed

by input retinal image. The input image will be compared with a validation check to verify that the necessary quality and format is achieved before any processing is performed by the system. This is a very important step because unacceptable or poor quality images can affect the accuracy of predictions. In case the image fails to pass the validation the image is discarded otherwise the workflow continues.

The image that has been validated is then preprocessed whereby it is standardized and enhanced. It includes sharpening, modifying the lighting and magnifying the picture. This is done to normalize the image such that the deep learning models can operate on the image. U-Net segmentation also then follows which is a significant process in detecting abnormalities in the retina. The model subdivides the image on the pixel level and identifies the locations of lesions. The presence of any lesions will be identified once the system is segmented. This choice is notable because it helps to reduce unnecessary calculations- in case of no lesion the system simply assigns the image in Grade 0 that indicates no disease. It is then subjected to the severity prediction phase, in case the lesions are confirmed and the level of diabetic retinopathy is determined. This move converts the visual data to a severity scale that is graded and is necessary to carry out additional clinical interpretation. Once classified, the system then proceeds to do clinical mapping using the UKPDS look up table that correlates DR grades with HbA1c ranges. The mean of this range is calculated to provide the approximate level of HbA1c. This step bridges the gap in the clinical and image-based analysis. This system then generates a feature table of a collection of features that is found on the HbA1c and the severity grade. These features are the inputs to the regression model, which forecasts the final level of blood glucose. The regression model is trained on real-world medical data (NHANES dataset) and prediction must be clinically relevant. Finally, the system then displays the resultant prediction and the process terminates at the end node. The diagram makes it easy to see the systematic manner in which the system includes validation, decision making and prediction to produce meaningful and accurate outputs.

3.3.7 Algorithm

The algorithm of the proposed system defines the step-by-step computational procedure followed by the pipeline to produce a final blood glucose prediction from a raw retinal fundus image. Starting from image preprocessing through Ben Graham correction and CLAHE, the algorithm sequentially passes the enhanced image through the U-Net segmentation model to generate a lesion mask, followed by the

ResNet-50 CNN for DR

grade classification using a 4-channel input. Predictions below the 0.45 confidence threshold are flagged as uncertain and not propagated further. The predicted grade is then mapped to an HbA1c midpoint using the UKPDS lookup table, from which a 7-feature vector is constructed and passed into the Lasso Regression model to produce the final fasting blood glucose prediction in mg/dL along with a risk classification and detected lesion summary.

Algorithm: *Diabetes Severity Prediction using Retinal Fundus Images*

Input: *Retinal fundus image I*

Output: *DR Grade, HbA1c, Blood Glucose (mg/dL), Risk Level, Detected Lesions*

Start

Step 1: Load retinal fundus image I

Step 2: Preprocessing

a. Remove black borders from I

b. Apply Ben Graham illumination correction

c. Apply CLAHE on LAB L-channel

d. Resize image to 512×512 → I_pre

Step 3: Lesion Segmentation

a. Pass I_pre through trained U-Net (EfficientNet-B4 encoder)

b. Generate 7-class lesion mask M

c. Identify detected lesion types from M
(Microaneurysms, Hemorrhages, Hard Exudates,
Soft Exudates, Neovascularization, IRMA)

Step 4: DR Grade Classification

a. Build 4-channel input X = [I_pre (RGB) + M]

b. Pass X through trained ResNet-50 CNN

c. Apply softmax → get grade probabilities

d. Predicted Grade G = class with highest probability

e. Confidence C = probability of predicted grade

f. If C < 0.45 → flag as Uncertain, stop
Else → proceed to Step 5

Step 5: HbA1c Estimation

a. Lookup UKPDS table using grade G:
G=0 → (5.5 – 6.5%) → midpoint 6.0

G=1 → (6.5 – 7.5%) → midpoint 7.0

G=2 → (7.5 – 8.5%) → midpoint 8.0

G=3 → (8.5 – 10.0%) → midpoint 9.25

G=4 → (10.0 – 12.0%)→ midpoint 11.0

b. Extract HbA1c midpoint

H_mid Step 6: Glucose Prediction

a. Construct 7-feature vector F:

F = [hba1c, hba1c_sq, hba1c_mid,

hba1c_range_w, grade_sq,

grade_hba1c, dr_grade]

b. Scale F using StandardScaler

c. Predict blood glucose:

BG = Lasso.predict(F_scaled)

Step 7: Assign Risk Level based on G:

G=0 → Low

G=1 →

Low-Moderate G=2

→ Moderate G=3 →

High

G=4 → Very High

Step 8: Display Final Output:

DR Grade: G

Confidence: C%

HbA1c Range: lo% – hi%

HbA1c Midpoint: H_mid%

Blood Glucose: BG mg/dL

Risk Level: risk_label

Detected Lesions: lesion_names

End

3.4 Requirements Engineering

Requirements engineering determines the functionality that the proposed system is to have, the system requirements and their limitations e.g. performance requirement. It ensures that they are addressed in the design and implementation stages of the system as such expectations exist between the technical and the user. The requirements have

been categorized into Functional Requirements and Non-Functional Requirements.

3.4.1 Functional Requirements

Functional requirements are concerned with the function of the system, its behavior, and features. The functional requirements of the system have been extracted from the proposed deep learning process which involves preprocessing of retinal image, segmentation of lesions, detection of diabetic retinopathy, and prediction of blood sugar level.

Image Upload and Validation

First, the software application should have the ability to upload the images in the commonly used file formats such as .jpg, .jpeg, and .png. Once the image is uploaded, the second step will be to validate the images in terms of their file format, resolution, and quality. Validation will help to prevent the processing of invalid images, which will affect the functioning of the system. Invalid images will prompt a warning message that the users need to submit other images. Some conditions need to be taken into consideration; for instance, when the user submits the file extension is incorrect. Moreover, the validation process will determine whether the image is suitable enough for the processing task, with an appropriate warning message.

Image Preprocessing

After the process of validation, the next step is automating the preprocessing of the system. The images need to be resized into a certain size, normalized intensity of pixel values, and contrast enhancement needs to be done. Using advanced methods of image processing such as CLAHE and Ben Graham preprocessing will be useful for improving the retinal structures. It is also important to consider denoising and illumination correction during the preprocessing phase to minimize any unwanted distortions in the input images. In the process of preprocessing the images, the most important anatomical parts such as blood vessels, optic disc, and macula should be prioritized, while the rest of the background parts are eliminated. There are several methods that can be implemented during the preprocessing phase such as extracting the regions of interest and circular cropping.

Preprocessing will also include application of artifact removal techniques to remove any unnecessary details such as camera reflections, image borders, or image noises that would negatively affect prediction accuracy of the model. Artifact removal techniques that can be used include histogram equalization and smoothing, provided that this is done without distorting important image details, especially

microaneurysms.

Lesion Segmentation

The model relies on a U-Net with an EfficientNet-B4 encoder for segmentation, which lets it zero in on both tiny details and the bigger picture in retinal images. It spots all sorts of trouble: microaneurysms, hemorrhages, hard and soft exudates, neovascularization, and those hard-to-catch IRMAs. When it processes an image, it generates a mask that marks exactly where each abnormality is on the retina. One big challenge is class imbalance. Some types of lesions barely show up, so it's easy for the model to overlook them. To fix that, it uses Dice loss and Focal loss. The classification part steps things up. Here, the model combines the raw retinal image with its lesion mask, stacking them as extra channels. This way, the network doesn't just see the overall picture—it gets a direct map of where things are going wrong. All this data gets passed into a ResNet-50, which then decides how severe the diabetic retinopathy is, with five possible grades from 0 to 4. To speed up learning, the model uses transfer learning, starting off with knowledge from networks already trained on other data. Since some severity grades don't show up as often, the training uses a weighted loss function so those rare cases don't get glossed over. Altogether, this approach makes the system much better at nailing down the exact stage of the disease.

Combination of Features and DR Severity Classification

Secondly, multi-channel input for the image and lesion mask will come in handy in enabling the network to learn common characteristics as well as characteristics that correspond to the lesions. The use of multi-channel input will enable classification of the images by the use of ResNet-50 classifier to determine the stage at which the retinopathy stands. It is important to observe that apart from the grade, a prediction is always accompanied by the degree of certainty attached.Transfer learning, label smoothing, and sampling weight will among many other approaches be considered when it comes to improving prediction capabilities. Firstly, the use of transfer learning in initializing the weights will help in fast convergence and effective feature learning from the input images. Label smoothing is important in ensuring that over-fitting does not occur in a bid to avoid over-representation of prediction. However, imbalanced labels may pose challenges in training, and such a challenge will be addressed using sampling weight. The presence of residual connection enables effective information flow within the network, hence helping to easily learn the characteristics irrespective of whether they are complex or not. Residual connection plays a vital role in addressing vanishing gradient problems in deep learning neural networks.

Mapping of Clinical Values and Features Engineering

Using the predictions, we build a lookup table that matches each DR severity grade with a specific HbA1c range. So, Grade 0 lines up with HbA1c levels between 5.5% and 6.5%. As the grade goes up, the HbA1c range gets higher too—Grade 4, for example, covers 10.0% to 12.0%. This gives us a straightforward link between how much the retina's been damaged and where the patient stands metabolically. Since HbA1c tracks average blood sugar over time, it's a solid clinical marker. To make things simpler, we use the middle value of the HbA1c range as a stand-in for each prediction. We don't stop there. We engineer more features: the raw HbA1c, its square ($HbA1c^2$), the calculated midpoint, and the full range. These additions pick up on both straight and not-so-straightforward patterns between HbA1c and glucose levels. More informative features mean we get a clearer snapshot of the patient's metabolic profile, which bumps up prediction accuracy. Before pushing this data through our regression model, we normalize and scale everything so things stay consistent.

Bringing in domain knowledge, we use these grade-to-range mappings to make the results easier to understand from a clinical perspective. This connects what the deep learning model spits out to real-world practice and makes our blood glucose predictions not just more trustworthy, but also more useful for doctors.

Regression-Based Model for Estimating Blood Glucose Levels

This phase will focus on the estimation of the blood glucose level by applying some of the regression models of machine learning including Lasso Regression, CART and Random Forest due to their capability of modelling both linear and non-linear relationships among variables. Training the models will be done by use of real datasets from the NHANES database. The inputs will comprise of the new engineered features HbA1c, HbA1c2, HbA1c Midpoint, HbA1c Range and Range Width. Performance assessment of the models will be made by considering some factors such as mean squared error (MSE), mean absolute error (MAE) and R-Squared (R2). Lasso Regression will assist in reducing the effects of insignificant variables, leading to a small model. Similarly, CART will assist in modelling non-linear relationships as well as explaining the model. Lastly, Random Forest improves the accuracy of the predictions. Model selection, which evaluates the performance of a model for its ability to make an accurate prediction will be applied during the selection of the best model. This will be followed by selection of the model, which will finally give the predicted value of the blood glucose level in mg/dL.

Presentation and Visualization of the Output.

The visualization and presentation of the output have to be user-friendly to ensure that non-technical individuals understand the results generated by the application. Amongst the results, there is the DR grade, confidence score, HbA1c range, estimated blood glucose level, and risk level which is categorized as low, medium, or high. All these variables together give a wholesome idea of the patient's condition. In addition to numbers, the application gives graphical representations such as the retinal image and the lesion overlay map, which are used to distinguish abnormalities and how the machine learning algorithm reached the decisions. Presentation includes proper formatting, labeling, color-coding based on the severity levels, and other graphical presentations, for instance, through charts. Graphical attributes can include the ability to zoom, pre-processing and segmented image comparison, and confidence level with the severity progress over time.

Model Performance Evaluation

In order to ensure that reliable and accurate results will be obtained, the system utilizes appropriate metrics in measuring the performance of the trained model. Accuracy, precision, recall, F1-score, and area under curve are just some of the metrics used in evaluating the classification algorithm in distinguishing the different stages of diabetic retinopathy. On the other hand, mean squared error, mean absolute error, and R-square are some of the metrics that are used to evaluate the ability of the algorithms to predict the value of blood glucose. For the validation of the results, cross-validation is used. In this way, the consistent performance of the classification algorithm when tested using different sets of data will be ensured. Lasso, CART, and Random Forest are among the models being compared in order to determine the best performing algorithm.

Reliability of the system and recovery.

The system must possess healthy error-handling to take care of the ad hoc situations such as invalid enticements, missing data or lack of processing. It should also ensure that any mistakes are handled gracefully, without causing any form of crashes or disruption of the system. Correct error messages should be displayed so that the user has a means of rectifying the errors to improve the overall usability**.**

3.4.2 Non-Functional Requirements

Non-Functional Requirements. Non-functional requirements will provide quality of system performance as well as the limits of operation in order to make it reliable, and

user friendly.

Performance

It should be ensured that the system performs extremely well not only in the case of classification of diabetic retinopathy but also in predicting the level of blood glucose. It should be noted that the performance of the classification of diabetic retinopathy is expected to be greater than acceptable; hence, this can be used in a healthcare setting. Besides, it should perform quickly by making predictions within just a few seconds. As far as the application of deep learning techniques like U-Net and ResNet-50 in the system is concerned, it will help avoid issues related to the performance of the system.

Scalability

Scalability is expected for the system in terms of its functionality, meaning that the system will effectively process large volumes of data, including thousands of retinal fundus images, without compromising its performance parameters. The volume and quality of the image will not affect the accuracy of the predictions made by the system. In designing a system that scales up the number of users, databases, and models, scalability should be considered. All changes must be incorporated into the existing architecture of the system.

Reliability

The results generated by the system should be accurate and reliable whenever a specific value is entered by a user. Any input values that are unexpected, including corrupt data or invalid file formats, should be managed by the system without crashing. Logs should be generated for error diagnosis.

Security and Privacy

Because the system involves health information, there are some aspects of security that should be taken into account. First of all, the data needs to be sent in a secure manner, and there should not be any leaks in any way. An authentication system should be provided as well. This kind of information is protected by the HIPAA and GDPR policy regulations. The data must remain private during the process of its collection, storage, and displaying.

Maintainability

The architecture of the system should be transparent and understandable by other developers. The modules that perform pre-processing, segmentation, classification, and prediction of the input data should be individually maintainable. All developers should write readable codes that can be changed by other members of the programming team without difficulty. New data and machine learning techniques

should be incorporated,

and it should be easy to change the pre-processing algorithm.

Portability

The system will operate on various operating systems including Windows, Linux, and Mac OS. In addition, the system will run on both local and cloud computing platforms to enhance portability. The system's library and dependency will not have any effect on the system's compatibility with various computing environments. The system needs low configurations for its deployment on different computing infrastructures.

Interoperability

The system will have the capability to interface with other healthcare management systems including HMS and PACS. Interoperability will utilize standardized formats and protocols. The system will support importation and exportation of retinal images in other health care management systems without affecting their operation.

Availability

There has to be a comprehensive system that provides for availability at a very high rate with the least downtimes possible to facilitate consistent utilization in both the clinical and research settings. The system would also provide for inherent mechanisms for overcoming challenges relating to the failure of either the hardware and software components of the AI systems, including the recovery processes involved. Maintenance should be planned to minimize any inconveniences caused. Availability would be essential for diagnostics and research.

Transparency and Explainability

The AI system should produce intelligible outputs to engender trust in the output of the AI system. Instead of offering black box solutions to problems, the system should produce visual explanations of the outputs, like placing lesions on the retinas. Through the visualizations provided, the physicians will understand the reasoning behind the output produced by the AI system.

3.5 Software Requirements and Hardware Requirements

Diabetes prediction using retinal images is a proposed AI diagnostic system that can be used to detect diabetes from the images captured using retinal imaging technology. The system involves a combination of hardware and software systems that aim at ensuring proper analysis, training, and visual presentation of results obtained using an AI model. The system involves the use of advanced AI algorithms that have been developed using Python software and are capable of handling massive amounts of data with the aid of

computer power. The algorithms involve processes such as preprocessing, lesion segmentation, classification, and prediction of blood glucose levels. Table 3 shows the list of software needed for implementation of the AI algorithm.

Table 3 Software Requirements

Serial No.	**Software**	**Purpose**
1	Operating System	A 64-bit Operating System such as Windows 10, Ubuntu 20.04+, or macOS provides a stable and compatible environment for running Python-based machine learning frameworks and ensuring efficient performance with hardware acceleration
2	Core Frameworks	
2.1	Python (3.8 or higher)	It is used as the main programming language in developing programs for preprocessing, segmentation, classification, and regression models. It has a strong community that makes use of the language for conducting medical image analysis.
2.2	TensorFlow/PyTorch	It is used in developing deep learning models for segmentation and classification using U-Net and ResNet-50, respectively. These platforms can utilize GPUs for fast model creation.
3	Key Libraries	
3.1	NumPy	It is used for numerical computations and manipulating arrays, especially during data preprocessing and model training.
3.2	Pandas	Contributes to processing structured data like NHANES dataset and creating feature tables in regression modeling.
3.3	Matplotlib	Helpful for visualization purposes, such as plotting model performance, learning curves, and results interpretation.
4	Image Processing	
4.1	OpenCV (cv2)	Can perform various pre-processing steps on images including resizing, filtering, CLAHE, and Ben Graham pre-processing techniques for retinal images

4.2	PIL(Python Imaging Library)	Used for image loading, conversion, and basic manipulation operations.
5	Machine Learning &Regression	
5.1	Scikit-learn	Used for implementing regression models such as Lasso, CART, and Random Forest for glucose prediction.
6	Development Environment	
6.1	KaggleNotebook	Interactive environments that run in the cloud can be used to program, test, and visualize models of deep learning. Kaggle provides access to huge datasets as well as GPU services to train the model without paying anything. Kaggle allows flexible execution with GPU/TPU services as well as easy integration with external storage services. Both can be used to develop models like U-Net and ResNet-50.

To achieve optimal performance during model training and inference, the key hardware configurations are outlined in Table 4

Table 4 Hardware Requirements

Serial No:	**Hardware**	**Purpose**
1	Processor (CPU)	The hardware requirements include a multi-core 64-bit processor like Intel Core i5/i7 or AMD Ryzen 5/7 to process and load data efficiently.
2	Accelerator (GPU)	A discrete graphics card that supports CUDA (like the NVIDIA GTX/RTX series with 4GB or 8GB of VRAM) will speed up processes such as segmentation and classification, leading to a reduction in the amount of time needed to train the mode

3	System Memory (RAM)	Minimum RAM requirement should be 8GB (more preferably 16GB or above), which is sufficient to store image data and other model parameters and perform processing tasks simultaneously.
4	Storage	At least 20GB or 30GB of free space in the hard drive is required to store the retinal dataset, trained model, feature table, and experimental results.
5	Display	High-definition display screen is essential to visualize the retinal images, lesion segmentation map, and result outputs for further analysis.

CHAPTER 4
RESULTS AND DISCUSSION

4.1 Dataset Description

Our project uses various clinically validated datasets that are well-established throughout its pipeline to make sure every step of processing is based on real-world medical data. The datasets utilized in this project include retinal imaging with fine-grained lesion annotations, clinical DR-to-HbA1c mappings, and large-scale epidemiological biochemical measurements which all play a unique and vital role in the overall system architecture. The judicious choice of these datasets makes the proposed model both technically and clinically significant since the sources of data are well-known in the medical research community. Collectively, these data sets compose the foundation of a multi-stage pipeline, starting with the raw retinal image analysis followed by the actual blood glucose estimation, rendering the system complete and suitable in the real-world application scenario, where diabetic patients are involved.

The initial and main dataset in this project is the Fine-Grained Annotated Diabetic Retinopathy (FGADR) dataset, where it will be used two times in two key steps of the project pipeline. The FGADR dataset has high-resolution retinal fundus image with pixel-level labeling of six different types of lesions, such as microaneurysms, hemorrhages, hard exudates, soft exudates, intraretinal microvascular abnormalities (IRMA), and neovascularization, and the DR severity grade label of each image. These pixel-level annotations, which are fine-grained, are employed in the first stage, and the U-Net segmentation model is trained with an EfficientNet-B4 encoder to be able to learn accurate lesion boundaries at different levels of diabetic retinopathy severity. The data offers a wide variety of levels of severity among its samples and hence, the segmentation model generalizes suitably across the wide variety of clinical manifestations, which is a vital feature of strong and credible lesion detecting performance.

At the second stage, the same FGADR data are reused in the process of DR grading by building a 4-channel input of the three RGB channels of the original retinal fundus image with the lesion segmentation mask created by the trained U-Net model. This 4-channel representation is used to train the ResNet-50 convolutional neural network, which predicts the DR severity grade on a scale from Grade 0 (No DR) to Grade 4

(Proliferative DR) as per the International Clinical Diabetic Retinopathy severity scale.

With the introduction of the segmentation mask as an extra input channel, valuable spatial information regarding the location of lesions is given to the ResNet-50 model, which greatly improves its capacity to separate between the adjacent grades of DR that can be visually similar. The two-fold use of FGADR dataset in both the segmentation and grading phase maintains consistency in the image distribution observed during the training phase, minimizing domain mismatch between the two models.

The latter data utilized in the present project is based on the United Kingdom Prospective Diabetes Study (UKPDS) clinical lookup tables that form a validated empirical correlation between the severity grades of DR and the HbA1c ranges that these grades are associated with. UKPDS is a clinical longitudinal study that followed thousands of diabetic patients over a number of years and carefully recorded the relationship between the long-term glycemic control as represented by levels of HbA1c and the development of diabetic retinopathy of all severity grades. The UKPDS tables employed in this project to predict the DR grade of the ResNet-50 model to a clinically validated HbA1c range, the midpoint HbA1c value of which is extracted and serves as a feature in the glucose estimation step. The rationale is that such a strategy provides the HbA1c estimation with no arbitrary reference but rather pegged on decades of longitudinal clinical data, giving the suggested non-invasive glucose prediction pipeline high medical merits.

The third data used in the present project is the National Health and Nutrition Examination Survey (NHANES) data set which is a large scale epidemiological survey conducted by the Centers for Disease Control and Prevention (CDC) that provides paired measurements of HbA1c and fasting blood glucose level among thousands of participants of various demographic groups. The statistical association between HbA1c and blood glucose levels was learned on NHANES dataset to produce a structured feature-engineered table, with Glucose as the target variable, having seven features, that is, hba1c, hba1c_sq, hba1c_mid, hba1c_range_w, grade_sq, grade_hba1c, and dr_grade. Such designed features not only capture linear but also non-linear characteristics of the HbA1c-glucose relationship, but the predictive models receive a rich and informative feature space to train on. The following table of feature values was used to train and assess four trained models: the ADA Formula baseline, Lasso Regression, CART, and Random Forest of which Lasso Regression had the lowest Mean Absolute Error (MAE), and was thus chosen as the

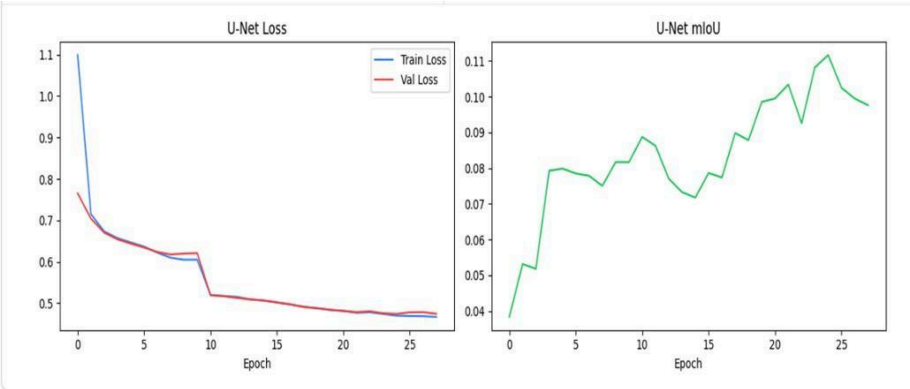
final glucose prediction model.

4.2 Experimental Results

The project is experimentally tested in three cascades of the pipeline, i.e., the lesion segmentation with the U-Net model, the DR grade classification with the ResNet-50 CNN, and the blood glucose prediction with the four regression models trained on the features acquired through NHANES. All the stages are measured with the help of relevant quantitative measures and presented in the form of graphs, confusion matrices, and a comparative bar chart to obtain a complete picture of the system functioning. The experiments are performed on the FGADR dataset using the stratified 70/15/15 train-validation-test split so that each DR grade class is equally represented in all of the three subsets, thus ensuring that no grade is over- or under-represented during evaluation. The whole pipeline is trained and tested on a Google PU-accelerated platform using PyTorch with CUDA support, such that the deep learning models can effectively converge within the allotted epochs.

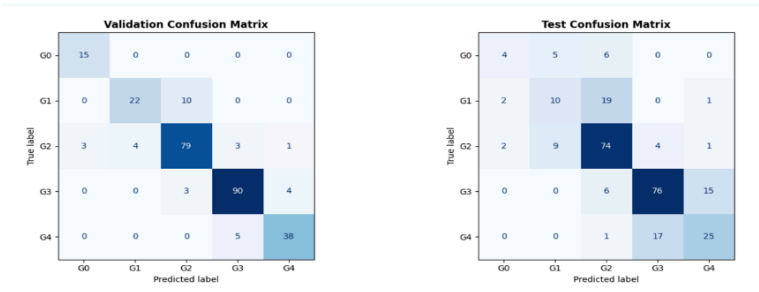
4.2.1 U-Net Lesion Segmentation Results

The U-Net model with EfficientNet-B4 encoder is trained over 28 epochs with a Dice and Focal loss function, which is chosen specifically to address the extreme imbalance in the number of background pixels versus lesion pixels in all six lesion types in the retinal lesion segmentation problem. The loss term of dice will clearly give the highest overlap between the predicted and ground truth lesion masks but the Focal loss term will depress the easy background pixels and force the model to pay attention to the hard lesion boundaries. The model is measured by the mean Intersection over Union (mIoU) as the main metric which is calculated on all the six different lesion classes without considering the background category to avoid overinflation of the score with the highest mIoU of 0.1116 recorded at epoch 25 as shown in Figure 8. The training and validation loss curves exhibit a steady decrease with each epoch, indicating the model learns meaningful lesion representations without overfitting, and the mIoU progression curve in Figure 10 indicates an upward trend with each epoch, with early stopping based on a patience of 7 epochs to avoid unnecessary computation once convergence is achieved. The last training loss and validation loss obtained after training are 0.4669 and 0.4745 respectively which show that there is a close agreement between the training and validation performance and further proves that there is no significant overfitting in the segmentation model.



4.2.2 ResNet-50 DR Grading Results

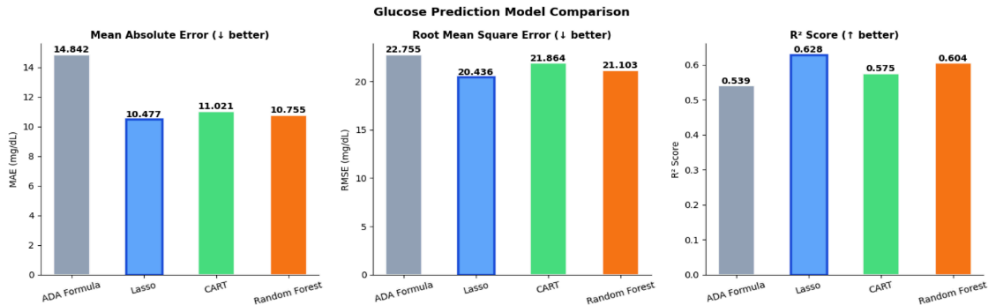
The DR grade classification ResNet-50 CNN is optimized over 23 epochs using a 4-channel input achieved by summing the three RGB channels of the preprocessed fundus image with a binary lesion segmentation mask produced by the trained U-Net model. The model is then trained with Cross Entropy loss and a Weighted Random Sampler is used on the training Data Loader to address the imbalance in the FGADR grade distribution, with some grades like Grade 0 and Grade 2 being more common than others. Performance on the held-out validation set and test set are measured individually and the model attains a validation performance of 88.09% and test performance of 68.23%. The confusion matrices created on both the validation and test splits, as shown in Figure 11 give a detailed, class-wise, breakdown of the correct and incorrect classifications of all five grade levels of DR between Grade 0 and Grade 4, and as may be seen, as shown in Figure 8, the model is able to classify the dominant grades as well as reveal grade adjacency errors where the model, in some instances, misclassifies vis The CNN output has a confidence thresholding mechanism with a threshold of 0.45 to indicate uncertain predictions so that any low-confidence grade prediction will not be passed into the downstream glucose estimation phase to avoid the overconfident low-quality prediction impacting the final glucose output.



4.2.3 Glucose Prediction Model Comparison

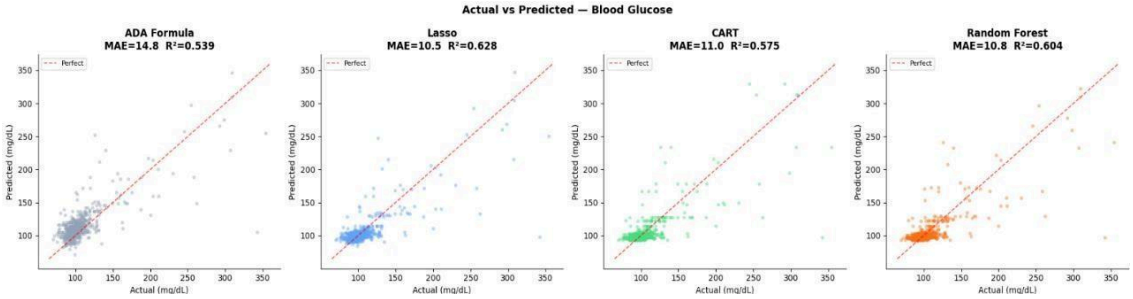
The glucose prediction phase is a training and comparison of four regression models on

the 7-feature NHANES-derived table with a 70/15/15-train-validation-test split, with the features of hba1c, hba1c_ sq, hba1c_mid, hba1c-range-w, grade- sq, grade-hba1c, and dr-grade as Four models were compared: ADA Formula baseline, Lasso Regression with 5-fold cross-validated alpha selection through Lasso CV, CART Decision Tree with Grid Search CV hyperparameter tuning and Random Forest Regressor, and compared on the same held-out test set using MAE, RMSE, and R2 Score to give a more complete picture of predictive accuracy, magnitude of error and variance explained. As Figure 12 demonstrates, the comparison of the bar charts of all three metrics clearly shows that Lasso Regression is the most successful model, as it has the lowest MAE of 10.477 mg/dL and the lowest RMSE of 20.436 mg/dL of all the four models and highest score in R 2 of 0.628, meaning that it accounts for about 62 The ADA Formula baseline has the weakest results in all three metrics with an MEA of 14.842 mg/dL, RMSE of 22.755 mg/dL, and R 2 of 0.539 which proves that the engineered 7-feature representation offers a much richer input space compared to the simple linear ADA formula on its own, and that the machine learning models trained. Further demonstrated by actual versus predicted blood glucose scatter plots in



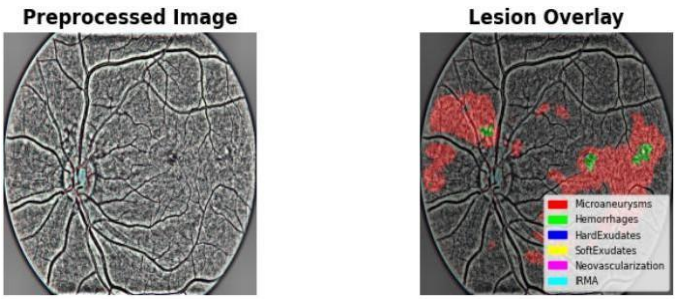
Figure

13, the Lasso Regression model results in prediction that are most tightly clustered around the perfect prediction diagonal (depicted as the red dashed line) throughout the entire range of glucose, especially in the lower range of 80 to 150 mg/dL where most NHANES samples fall. Random Forest model has the second best MAE of 10.755 mg/dL and R 2 of 0.604, close behind CART which has an MAE of 11.021 mg/dl and R 2 of 0.575, and much narrower spread around the perfect prediction line in Figure 10 than Lasso, especially at high glucose levels above 200 Considering the overall comparison of all three evaluation metrics and the graphic representation of the scatter plots in Figure 11, the final glucose prediction model to be incorporated in the proposed pipeline is Lasso Regression, and the results of its predictions are taken as the final output of the diabetes severity prediction system.

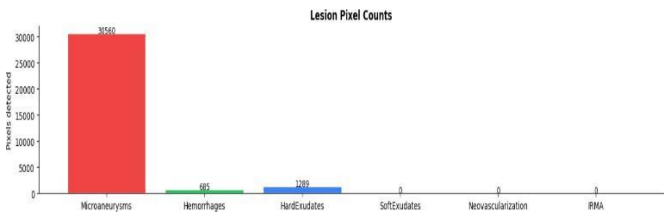


4.2.4 Final Pipeline Output

A full end-to-end pipeline is tested by processing a sample retinal fundus image through the five stages and ensuring it passes through all stages to ensure that pipeline elements work as expected and generate meaningful clinical results. Figure 14 shows the processed retinal fundus image after Ben Graham correction and CLAHE enhancement with the lesion overlay generated by the trained U-Net segmentation model, with the color-coded overlay indicating clearly the location of the Microaneurysms and Hemorrhages as the two main types of lesions found in the sample image.



As shown in Figure 15 of the bar chart of the lesion pixel count, Microaneurysms have the highest pixel count of 41,830 detected pixels, with the next pixel counts of 1,941, 15, and 11 pixels of Hard Exudates and Soft Exudates respectively, and zero pixels of Neovascularization and IRMA in this specific sample. The prevalence of Microaneurysms and Hemorrhages in the distribution of lesions is also clinically consistent with the advanced diabetic retinopathy, which further confirms the consistency of the segmentation output with the downstream DR grade prediction.



The bar chart of grade probabilities generated by the ResNet-50 CNN as shown in Figure 16 puts the grade of 4 (Proliferative DR) and Grade 3 (Severe DR) at 91.5 and 8.2 percent respectively and all other grades are assigned a value of 0.00. The confidence of 91.5% is significantly above the uncertainty threshold of 45% represented by the red dashed line in Figure 16 and so the model is very confident in its Grade 4 prediction of this sample and the prediction is not regarded as uncertain and thus passes directly into the downstream HbA1c estimation step without an uncertainty flag.

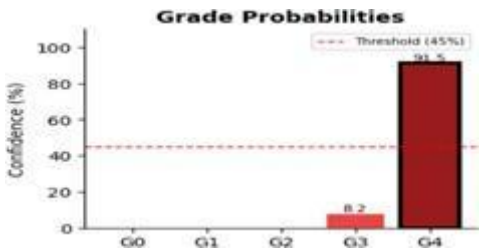


Figure 16 Grade Probabilities

The predicted DR Grade 4 is mapped to an HbA1c range of 10.0% to 12.0% as shown in Table 5 using the UKPDS lookup table where the midpoint HbA1c of 11.0% is obtained and fed into the 7-feature vector to predict glucose. Lasso regression model, the best model according to the lowest MAE, yields a final predicted VH-risk-based fasting blood glucose level of 263.5 mg/dL, which is consistent with the Proliferative DR grade and high range of HbA1c, corroborating the validity of all five stages of the proposed pipeline as one that integrates effectively to yield consistently and clinically meaningful results.

Table 5 Prediction Summary

DR Grade	4
Diagnosis	Proliferative DR
confidence	91:5%
Hb1Ac range	10.0-12.0%
Hb1Ac mid	11.0%
Glucose	263.5 mg/dL
Risk	Very High
Model Used	Lasso

4.3 Advantages of Proposed Work

The suggested diabetes severity prediction system based on retinal fundus images is a solution to a long-standing and critical problem with diabetes mellitus management, that is, the non-invasive and affordable estimation of blood glucose and the severity of diabetic retinopathy without the necessity of any biochemical blood assays and laboratory equipment. Diabetes is among the most common chronic illnesses in the world and early diagnosing and regular check-ups are imperative to avoid life-threatening complications, including blindness, kidney failure, and cardiovascular disease. The proposed system has a direct connection to this requirement by providing a completely automated image-based diagnostic pipeline that can be launched in resource-constrained clinical environments where regular testing of HbA1c and ophthalmological grading by trained experts might not be easily accessible. The proposed work provides new opportunities in the large-scale diabetic screening program both in urban and rural healthcare settings by filling the gap between retinal imaging and systemic glycemic estimation.

The main beneficiaries of the proposed system are diabetic patients, especially those in low- and middle-income environments who do not have access to frequent laboratory tests and expert ophthalmological assessments. This system would be of great use to general practitioners and primary care doctors as well since it would give them an automated second opinion regarding the severity of diabetic retinopathy and an approximate blood sugar level to guide a more informed clinical decision without necessarily referring to a specialist in each instance. Also, mass public health education and diabetic screening clinics can benefit the proposed work as the system can be fed with a retinal fundus image only and processed completely using software, so it can be implemented on regular computing hardware without any special medical labs. This pipeline can also be integrated on the telemedicine platforms and mobile health applications to reach geographically remote populations that are not otherwise adequately served by the current healthcare system.

From a technical standpoint, the proposed system offers several meaningful advantages over existing approaches in the literature. The use of Ben Graham preprocessing combined with CLAHE enhances image quality by correcting uneven illumination and improving local contrast in the retinal fundus images, which directly improves the ability of downstream models to detect subtle lesion structures that are often missed in

raw, unprocessed images. The U-Net architecture with an EfficientNet-B4 encoder provides a powerful and parameter-efficient segmentation backbone that is well-suited for the fine-grained, pixel-level lesion annotation task required by the FGADR dataset, and the use of a combined Dice and Focal loss function specifically addresses the severe class imbalance between lesion and background pixels that is a well-known challenge in medical image segmentation tasks. The 4-channel input design of the ResNet-50 CNN, which incorporates the segmentation mask alongside the original RGB image, provides the grading model with explicit spatial lesion context that purely image-based grading approaches lack, resulting in more accurate and interpretable grade predictions across all five severity levels.

Technically, the proposed system has a number of significant benefits over other methods in the literature. Ben Graham preprocessing with CLAHE improves the quality of the retina fundus images by equalizing uneven illumination and improving local contrast in the retinal fundus images which directly increases the capability of downstream models to identify subtle lesion structures that are commonly missed in unprocessed images. The U-Net architecture with an EfficientNet-B4 encoder provides a powerful and parameter-efficient segmentation backbone that is well-suited for the fine-grained, pixel-level lesion annotation task required by the FGADR dataset, and the use of a combined Dice and Focal loss function specifically addresses the severe class imbalance between lesion and background pixels that is a well-known challenge in medical image segmentation tasks. The 4-channel input design of the ResNet-50 CNN that includes the segmentation mask along with the original RGB image gives the grading model explicit spatial lesion information that the methods based on images do not, leading to more predictive and interpretable grade predictions at all five severity levels.

The medically based two-stage bio-chemical estimation pipeline, with the DR grades mapped to the HbA1c ranges using the clinically validated UKPDS tables and the Lasso regression trained to predict the fasting blood glucose using the NHANES data, is a medically based approach that makes sure that the final glucose estimate is not arbitrary but rather tied to decades of epidemiological and clinical data. The feature engineering step, which builds seven interpretable features revealing linear and non-linear HbA1c-glucose associations, guarantees that the Lasso regression model is allowed access to a highly-informed feature space, yet is computationally lightweight and interpretable, which is a major strength in clinical settings where physician trust

to the model and

regulatory compliance require feature transparency.

4.4 Limitations

Although the proposed retinal fundus-based diabetes severity prediction system has a variety of technical and clinical contributions, it has multiple inherent limitations that are caused by the type of the datasets used, assumptions that are incorporated in the pipeline design, and limitations of the work of individual models that are used at each step. Proper knowledge of these limitations is needed not only to accurately interpret the results described in this project, but also to see the ways in which the system needs to be refined before it can be deemed to be sufficiently ready to be implemented in real-life clinical settings. These constraints cut across the five phases of the pipeline, and are elaborated in this section to give a sincere and thorough evaluation of the constraint limits of the proposed work.

The initial and the most important limitation of the suggested system is the fact that it solely uses the FGADR dataset to train lesion segmentation model and DR grading model. Although FGADR is a highly annotated benchmark, it is a limited set of retinal images with respect to patient demographics, imaging devices, and clinical environments, i.e. the trained models might not generalize to fundus images taken by other cameras, under varying lighting conditions, or on patients with varying ethnic or physiological profiles. This problem of domain shift restricts the practical strength and clinical transferability of the segmentation and grading models to images in hospitals or screening programs with different imaging protocols as the ones used to construct the FGADR dataset.

The second important drawback is the step of the HbA1c estimation, which is based on the UKPDS lookup table to interpolate a predicted DR grade and a fixed range of HbA1c to extract the midpoint as a single point estimate. The method is simplistic because the association between the level of DR severity and the HbA1c level is very complex and highly individual, with the same level of DR severity producing a broad spectrum of HbA1c levels in different patients due to various factors such as disease duration, treatment history, and individual variability in metabolism. The system ignores all the variance within a grade, and assumes that all patients with a particular DR grade would have exactly the same level of HbA1c, which is clinically implausible and can cause systematic error in the final glucose prediction step.

The third drawback is related to the glucose prediction models that are trained on the

NHANES dataset, in which the DR grade feature utilized during training is not directly predicted by a CNN grading model but is rather estimated approximately using the values of HbA1c in a rule-based mapping because the NHANES dataset does not include actual DR grade annotations. This difference between training-time feature construction based on reconstructed approximate grades and inference-time feature construction based on real CNN-predicted grades causes a train-test distribution mismatch, which can adversely affect the quality of glucose predictions especially in patients whose CNN-predicted DR grade is not the same as the grade that would have been reconstructed given their actual HbA1c value.

The fourth limitation is associated with the general sequential pipeline model, in which any errors introduced at one point are propagated unconditionally to all the following points, without a means to correct or quantify this uncertainty. A false prediction of the DR grade by the ResNet-50 CNN will also cause a false prediction of the UKPDS HbA1c range look-up, which will then cause a false glucose estimate to be introduced into the glucose prediction model, ultimately causing a glucose error that is biased in the same direction as the upstream grading error. A lack of uncertainty propagation between stages implies that the system will only give a single point estimate of glucose without any confidence interval associated with it, which poses a great constraint in a clinical setting where a physician must verify how reliable a prediction is prior to making any diagnostic or treatment judgments based on it.

CHAPTER 5
CONCLUSION AND FUTURE WORK

5.1 Conclusion

The proposed system has been effective in meeting the urgent requirement of a non-invasive, automated substitute to the traditional blood glucose testing as it shows that retinal fundus images are also a feasible proxy to estimate systemic glycemic levels in diabetic patients. The pipeline follows five clearly defined steps, namely, retinal image preprocessing, U-Net based lesion segmentation, ResNet-50 based DR grade classification, UKPDS-driven HbA1c mapping, and NHANES based Lasso regression to predict glucose, which are supported by relevant clinical evidence and are measured using relevant performance metrics.

The proposed system has three highly selected and clinically verified sources as its dataset base, each performing a unique purpose in the pipeline. The Fine-Grained Annotated Diabetic Retinopathy (FGADR) dataset which has 1842 retinal fundus images with pixel-level lesion annotations in six lesion types and DR grade labels has a dual purpose of providing training data to both the U-Net segmentation model and the ResNet-50 DR grading model, with a stratified train-validation-test split (70/15 The UKPDS clinical lookup tables of prospective diabetes study offer the clinically validated bridging between the predicted DR grades and the actual HbA1c ranges that anchor the biochemical estimation phase in decades of longitudinal clinical evidence. The National Health and Nutrition Examination Survey (NHANES) data provides paired HbA1c and fasting blood glucose values in thousands of participants with diverse demographics, and on which the glucose regression models are trained and tested.

It has a best mIoU of 0.1116 on segmented lesions of six classes and a validation accuracy of 88.09% on DR grade on five severity levels, which proves that the deep learning parts are effective at capturing the retinal appearance of diabetic progression. Out of the four regression models tested on NHANES-derived features, Lasso Regression is the most efficient with a lowest MAE of 10.477 mg/dL and highest R 2 of 0.6283 and will thus be chosen as the final glucose prediction model. The end-to-end coherence of the pipeline across all five stages is verified by the fact that the sample result of 263.5 mg/dL in a Grade 4 Proliferative DR image with a CNN confidence of

91.5 shows that the detected lesions of Microaneurysms and Hemorrhages are fully

aligned with the predicted severe DR grade.

The main originality of the work is the pioneering combination of deep learning-based retinal image analysis with the traditional clinical maps, namely the UKPDS DR-to-HbA1c lookups and NHANES HbA1c-glucose data, creating a medically-grounded interpolation of retinal results to blood glucose estimation. The Ben Graham illumination correction and CLAHE preprocessing pipeline ensures that the retinal images presented to the deep learning models are of similar quality in all cases, even in cases where the imaging conditions were not optimal, whereas the 4-channel input format of the ResNet-50 CNN, which uses the U-Net generated lesion mask in addition to the RGB image, gives the grading model explicit spatial lesion information that greatly improves grade The feature engineering step, which builds seven interpretable features that model both a linear and a non-linear relationship between HbA1c and glucose, provides the Lasso regression model with access to a rich and informative feature space with a critical benefit in the clinical setting as the transparency of the model is critical to physician trust.

The suggested system has a high potential of implementation in resource-constrained healthcare systems, telemedicine applications, and large-scale diabetic screening programs, where it can be utilized to provide the opportunity to identify and monitor the severity of diabetes in the now underserved population with respect to traditional diagnostic tools. The system enables diabetes severity evaluation to be more approachable, cost-effective, and scalable by reducing the reliance on invasive blood samples and laboratory infrastructure to a single retinal fundus photograph, which is a significant initiative to democratize the process of diabetic screening and make non-invasive glycemic monitoring accessible to populations that need it the most.

5.2 Future Enhancements

While the proposed system has demonstrated encouraging results in all five stages of the proposed pipeline, there are several ways the system can be enhanced and improved to overcome the current limitations and to further improve its clinical relevance and practicality.

The first and most significant future enhancement is to build a multi-modal dataset of retinal fundus images, DR grades, HbA1c and fasting blood glucose of the same patients at the same time. This would remove the train-test distribution shift in the glucose prediction stage where DR grade features are substituted by HbA1c values

during training when DR grade labels are not available in the NHANES dataset, and would greatly improve the accuracy and acceptability of the final glucose prediction.

The second improvement is to replace the UKPDS midpoint extraction with a probabilistic model that converts the DR grade predictions into a distribution over HbA1c values, rather than a single point estimate, so that uncertainties of the DR grade predictions would be propagated through the pipeline and clinicians would be provided with glucose prediction intervals, which would significantly boost their confidence in the system's output.

The third enhancement is to extend the current system to predict diabetes type (Type 1 or Type 2) based on the predicted DR grade, HbA1c level and blood glucose value, given the different retinal presentation and glucose progression patterns associated with Type 1 and Type 2 diabetes. Using additional information about the onset of diabetes, indicators of insulin dependency, and C-peptide in combination with the retinal diagnosis would allow the system to infer between the two types of diabetes and enhance the system's diagnostic coverage and clinical utility beyond the grade.

The fourth improvement is to adapt the lesion segmentation model to identify other retinal pathologies such as diabetic macular edema and optic disc anomalies using datasets such as IDRiD or MESSIDOR-2 that would provide more informative lesion features to the DR grading and glucose prediction stages, and would allow a more comprehensive retinal analysis.

Finally, the system can be run as a fully cloud-based, real-time diagnostic service on Amazon Web Services (AWS) with a mobile-optimized user interface that would enable clinicians and screening program managers to upload fundus images and receive real-time diagnostic results on their mobile devices, thus extending the proposed system to remote and resource-scarce populations currently lacking access to traditional diagnostic approaches.

CHAPTER 6 APPENDICES

Sample Code :

```
import os import cv2 import
warnings
import numpy as np import joblib
from pathlib import Path

import torch
import torch.nn as nn
import torchvision.models as models
import torchvision.transforms as T

import segmentation_models_pytorch as smp
import albumentations as A
from albumentations.pytorch import ToTensorV2

warnings.filterwarnings('ignore')
# CONFIG
DEVICE        = 'cuda' if torch.cuda.is_available() else 'cpu'
IMG_SIZE      = 512
CONF_THRESH = 0.45
MODEL_DIR = Path(_file_).parent / 'models'
LESION_NAMES = {
        1: 'Microaneurysms',
        2: 'Hemorrhages',
        3: 'HardExudates',
```

4: 'SoftExudates',

5: 'Neovascularization',

6: 'IRMA',

```
}
LESION_COLORS = {
    0: [0,  0,  0],
    1: [255, 0,  0],
    2: [0,  255, 0],
    3: [0,  0,  255],
    4: [255, 255, 0],
    5: [255, 0,  255],
    6: [0,  255, 255],
}
DR_TO_HBA1C = {
    0: {'range': (5.5, 6.5), 'label': 'No DR',          'risk': 'Low',          'color':
'#22C55E'},
    1: {'range': (6.5, 7.5), 'label': 'Mild DR',          'risk': 'Low-Moderate', 'color':
'#EAB308'},
    2: {'range': (7.5, 8.5), 'label': 'Moderate DR',     'risk': 'Moderate', 'color':
'#F97316'},
    3: {'range': (8.5, 10.0), 'label': 'Severe DR',          'risk':                'High',
    'color': '#EF4444'},
    4: {'range': (10.0, 12.0), 'label': 'Proliferative DR', 'risk': 'Very High',     'color':
'#991B1B'},
}
FEATURE_NAMES = ['hba1c', 'hba1c_sq', 'hba1c_mid', 'hba1c_range_w', 'grade_sq',
'grade_hba1c', 'dr_grade']

# MODULE 1 — PREPROCESSING

def ben_graham(image, sigma=10):
    """Removes uneven illumination."""
    return cv2.addWeighted(image, 4, cv2.GaussianBlur(image, (0, 0), sigma), -4,
128)

def apply_clahe(image, clip_limit=3.0, tile_size=8):
    """CLAHE on LAB L-channel."""
```

```
    lab = cv2.cvtColor(image, cv2.COLOR_RGB2LAB)
    clahe = cv2.createCLAHE(clipLimit=clip_limit,
tileGridSize=(tile_size, tile_size))
    lab[:, :, 0] = clahe.apply(lab[:, :, 0])
    return cv2.cvtColor(lab, cv2.COLOR_LAB2RGB)

def remove_black_border(image):
    gray = cv2.cvtColor(image, cv2.COLOR_RGB2GRAY)
    _, mask = cv2.threshold(gray, 10, 255, cv2.THRESH_BINARY)
    coords = cv2.findNonZero(mask)
    if coords is None:
    return image
    x, y, w, h = cv2.boundingRect(coords)
    return image[y:y + h, x:x + w]

def preprocess_fundus(image, target_size=512):
    image = remove_black_border(image)
    image = ben_graham(image)
    image = apply_clahe(image)
    image = cv2.resize(image, (target_size, target_size))
    return np.clip(image, 0, 255).astype(np.uint8)

# MODULE 2 — U-NET MODEL DEFINITION

def build_unet():
    return smp.Unet(
    encoder_name='efficientnet-b4',
    encoder_weights=None,      # weights loaded from file, not re-downloaded
    in_channels=3,
    classes=7,
    activation=None,
    ).to(DEVICE)

# MODULE 3 — CNN MODEL DEFINITION (ResNet-50, 4-channel)
```

```
class DRGradeCNN(nn.Module):
    """
    ResNet-50 fine-tuned for DR grade classification.
    conv1 modified: 3-channel → 4-channel (RGB + lesion mask).
    Custom head: FC(2048→512) → BN → ReLU → Dropout → FC(512→256)
→ FC(256→5)
    """
    def _init_(self, num_classes=5, dropout=0.45):
    super()._init_()
    backbone      = models.resnet50(weights=None)
    old_conv      = backbone.conv1
    backbone.conv1 = nn.Conv2d(4, 64, kernel_size=7, stride=2, padding=3,
bias=False)
    with torch.no_grad():
    backbone.conv1.weight[:, :3] = old_conv.weight
    backbone.conv1.weight[:, 3] = old_conv.weight.mean(dim=1)
    self.features = nn.Sequential(*list(backbone.children())[:-1])
    self.classifier = nn.Sequential(
    nn.Flatten(),
    nn.Linear(2048, 512), nn.BatchNorm1d(512), nn.ReLU(inplace=True),
nn.Dropout(dropout),
    nn.Linear(512, 256), nn.BatchNorm1d(256), nn.ReLU(inplace=True),
nn.Dropout(dropout * 0.6),
    nn.Linear(256, num_classes),
    )

    def forward(self, x):
    return self.classifier(self.features(x))
```

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